

Electrophysiological dynamics of a triple network model of cognitive control: A multi-experiment replication

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Abstract

Dynamic interactions between large-scale brain networks are thought to underpin human cognitive processes, but their underlying electrophysiological dynamics remain unknown. The triple network model, which highlights the salience, default mode, and frontoparietal networks, provides a fundamental framework for understanding these interactions. To unravel the electrophysiological mechanisms underlying these network dynamics, we utilized intracranial EEG recordings from 177 participants across four distinct memory experiments. Our findings revealed a consistent pattern of directed information flow from the anterior insula, a key node of the salience network, to both the default mode and frontoparietal networks. Notably, this pattern of information transmission was observed regardless of the nature of the tasks, whether they involved externally driven stimuli during encoding or internally governed processes during free recall. Moreover, the directed information flow from the anterior insula to the other networks was present irrespective of the activation or suppression states of individual network nodes. Furthermore, we observed a specific suppression of high-gamma power in the posterior cingulate cortex/precuneus node of the default mode network during memory encoding, but not recall, suggesting a task-specific functional down-regulation of this region. Crucially, these results were reliably replicated across all four experiments, underscoring the robustness and generalizability of our findings. Our study significantly advances the understanding of how coordinated neural network interactions underpin cognitive operations and highlights the critical role of the anterior insula in orchestrating the dynamics of large-scale brain networks. These findings have important implications for elucidating the neural basis of cognitive control and its potential disruptions in various neurological and psychiatric disorders.

eLife assessment

In this manuscript, the authors present **valuable** findings on the apparent role of a salience-network anterior insula node in directing fronto-parietal and default-mode network activity within a tripartite network during control of memory, drawn from an impressive invasive human neurophysiological dataset. While we commend the use of a large intracranial EEG dataset to approach this question, the study at present is **incomplete** in its methodologies, analysis, and interpretation to support the authors' central claims. The manuscript could be improved by addressing the concerns described.

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Introduction

Dynamic interactions between large-scale brain networks are thought to underpin human cognitive processes, but the electrophysiological dynamics that underlie these interactions remain elusive. The triple network model, which includes the salience (SN), default mode (DMN), and frontoparietal networks (FPN), offers a fundamental framework for understanding these complex interactions (Cai, Ryali, Pasumarthy, Talasila, & Menon, 2021 [↗](#); Menon, 2011 [↗](#), 2023 [↗](#)). These networks collaboratively manage tasks that require significant cognitive control, highlighting the integrated nature of brain function. Building on Mesulam's (Mesulam, 1990 [↗](#)) theory that all cognitive and memory systems operate within a complex architecture of interconnected brain regions, the triple network model articulates how these networks facilitate demanding cognitive tasks. However, despite the model's broad influence, the specific electrophysiological mechanisms that support these interactions during cognitive tasks remain poorly understood.

The SN, DMN, and FPN each play unique and critical roles in modulating human cognition (Menon, 2023 [↗](#)). The SN, known for its role in identifying and filtering salient stimuli, helps individuals focus on goal-relevant aspects of their environment or task, thus enhancing engagement with task-relevant stimuli (Menon & Uddin, 2010 [↗](#)). The DMN is typically active during internally focused cognitive processes and is implicated in various memory functions, such as the retrieval of past experiences, envisioning future events, and forming detailed episodic memory traces (Buckner, Andrews-Hanna, & Schacter, 2008 [↗](#); Fox & Raichle, 2007 [↗](#); Fox et al., 2005 [↗](#); Greicius et al., 2008 [↗](#); Greicius & Menon, 2004 [↗](#); Laufs et al., 2003 [↗](#); Raichle, 2015 [↗](#); Raichle et al., 2001 [↗](#); Smallwood et al., 2021 [↗](#)). Conversely, the FPN is involved in the maintenance and manipulation of information within working memory and exerts top-down cognitive control during memory formation (Badre, Poldrack, Paré-Blagoev, Insler, & Wagner, 2005 [↗](#); Badre & Wagner, 2007 [↗](#); Helfrich & Knight, 2016 [↗](#); Jin, Oik, & Hilgetag, 2010 [↗](#); Simons & Spiers, 2003 [↗](#); Uncapher & Wagner, 2009 [↗](#); Wagner, Paré-Blagoev, Clark, & Poldrack, 2001 [↗](#); Wagner, Shannon, Kahn, & Buckner, 2005 [↗](#)).

Central to the functionality of this model is the anterior insula (AI), a pivotal node within the SN. Human functional magnetic resonance imaging studies have revealed a critical role for the SN in regulating the engagement and disengagement of the DMN and FPN across diverse cognitive and affective tasks (Bressler & Menon, 2010 [↗](#); Cai et al., 2016 [↗](#); Cai et al., 2021 [↗](#); Chen, Cai, Ryali, Supekar, & Menon, 2016 [↗](#); Kronemer et al., 2022 [↗](#); Raichle et al., 2001 [↗](#); Seeley et al., 2007 [↗](#); Sridharan, Levitin, & Menon, 2008 [↗](#)). The AI dynamically detects and filters task-relevant information, facilitating rapid and efficient switching between the DMN and FPN in response to

shifting task demands (Menon, 2015 [↗](#)). Yet, how this process operates at the neurophysiological level remains unknown, underlining a significant gap in our understanding of the functioning of cognitive control networks.

Episodic memory, the cognitive process of encoding, storing, and retrieving personally experienced events, is essential for a variety of complex cognitive functions and everyday activities (Dickerson & Eichenbaum, 2010 [↗](#); Düzel, Penny, & Burgess, 2010 [↗](#); Moscovitch, Cabeza, Winocur, & Nadel, 2016 [↗](#); Ranganath & Ritchey, 2012 [↗](#); Rugg & Vilberg, 2013 [↗](#); Rutishauser, Reddy, Mormann, & Sarntin, 2021 [↗](#); Yonelinas, Ranganath, Ekstrom, & Wiltgen, 2019 [↗](#)). Influential theoretical models of human memory posit a key role for control processes in regulating hierarchical processes associated with episodic memory formation (Andermane, Joensen, & Horner, 2021 [↗](#); Atkinson & Shiffrin, 1968 [↗](#); Bastos et al., 2012 [↗](#); Kumaran & McClelland, 2012 [↗](#); Tulving, 2002 [↗](#)). The formation of episodic memories relies on the intricate interplay between large-scale brain networks, making it an ideal cognitive process to investigate the triple network model's applicability and the underlying neurophysiological dynamics. Understanding the neurophysiological dynamics that support episodic memory is essential not only for elucidating basic brain functions but also for addressing neuropsychological disorders where these mechanisms may be disrupted (Dickerson & Eichenbaum, 2010 [↗](#); Grady, Furey, Pietrini, Horwitz, & Rapoport, 2001 [↗](#); Uhlhaas & Singer, 2006 [↗](#)).

Our understanding of the dynamic network interactions during human cognition, especially those involving large-scale brain networks, is primarily informed by fMRI studies (Menon, 2011 [↗](#); Uddin, 2015 [↗](#)). However, the temporal resolution of these studies, typically around two seconds, is a significant constraint. This impedes our understanding of real-time, millisecond-scale neural dynamics and underscores the need to explore these networks' interactions at time scales more pertinent to neural circuit dynamics. The difficulties involved in acquiring intracranial EEG (iEEG) data from multiple brain regions have made it challenging to elucidate the precise neural mechanisms underlying the functioning of large-scale networks. These challenges obscure our understanding of the dynamic temporal properties and causal interactions between the AI and other large-scale distributed networks during memory formation.

To address these gaps, we leveraged iEEG data acquired during multiple memory experiments from the University of Pennsylvania Restoring Active Memory (UPENN-RAM) study (Solomon et al., 2019 [↗](#)). This dataset offers an unprecedented opportunity to probe the electrophysiological dynamics of triple network interactions during episodic memory formation, with depth recordings from 177 participants across multiple memory experiments. The UPENN-RAM dataset includes electrodes in the AI, the posterior cingulate cortex (PCC)/precuneus and medial prefrontal cortex (mPFC) nodes of the DMN, and the dorsal posterior parietal cortex (dPPC) and middle frontal gyrus (MFG) nodes of the FPN. By leveraging this large-scale dataset, we sought to elucidate the neurophysiological underpinnings of the AI's dynamic network interactions with the DMN and FPN and to assess the consistency of these interactions across diverse memory tasks, thus enhancing the robustness and reliability of our findings.

We investigated four episodic memory experiments spanning both verbal and spatial domains. The first experiment was a verbal free recall memory task (VFR) in which participants were presented with a sequence of words during the encoding period and asked to remember them for subsequent verbal recall. The second was a categorized verbal free recall task (CATVFR) in which participants were presented with a sequence of categorized words during the encoding period and asked to remember them for subsequent verbal recall. The third involved a paired associates learning verbal cued recall task (PALVCR) in which participants were presented with a sequence of word-pairs during the encoding period and asked to remember them for subsequent verbal cued recall. The fourth was a water maze spatial memory task (WMSM) in which participants were shown objects in various locations during the encoding periods and asked to retrieve the location

of the objects during a subsequent recall period. This comprehensive approach afforded a rare opportunity in an iEEG setting to examine network interactions between the AI and the DMN and FPN nodes during both encoding and recall phases across multiple memory domains.

A crucial test of the triple network model is whether the AI exerts a strong directed causal influence on the DMN and FPN. The AI is consistently engaged during attentional tasks and, dynamic causal modeling of fMRI data suggests that it exerts strong causal influences on the DMN and FPN in these contexts (Cai et al., 2016 [DOI](#); Cai et al., 2021 [DOI](#); Chen et al., 2016 [DOI](#); Sridharan et al., 2008 [DOI](#); Wen, Liu, Yao, & Ding, 2013 [DOI](#)). However, it remains unknown whether the AI plays a causal role during memory formation and whether such influences have a neurophysiological basis. To investigate the directionality of information flow between neural signals in the AI and DMN and FPN, we employed phase transfer entropy (PTE), a robust and powerful measure for characterizing information flow between brain regions based on phase coupling (Hillebrand et al., 2016 [DOI](#); Lobier, Siebenhüner, Palva, & Matias, 2014 [DOI](#); Wang et al., 2017 [DOI](#)). Crucially, it captures linear and nonlinear intermittent and nonstationary causal dynamics in iEEG data (Hillebrand et al., 2016 [DOI](#); Lobier et al., 2014 [DOI](#); Menon et al., 1996 [DOI](#)). We hypothesized that the AI would exert higher directed causal influence on the DMN and FPN than the reverse.

To further enhance our understanding of the dynamic activations within the three networks during episodic memory formation, we determined whether high-gamma band power in the AI, DMN, and FPN nodes depends on the phase of memory formation. Memory encoding, driven primarily by external stimulation, might invoke different neural responses compared to memory recall, which is more internally driven (Andrews-Hanna, 2012 [DOI](#); Buckner et al., 2008 [DOI](#)). We hypothesized that DMN power would be suppressed during memory encoding as it is primarily driven by external stimuli, whereas an opposite pattern would be observed during memory recall which is more internally driven. Based on the distinct functions of the DMN and FPN—internally-oriented cognition and adaptive external response—we expected to observe differential modulations during encoding and recall phases. By testing these hypotheses, we aimed to provide a more detailed understanding of the dynamic role of triple network interactions in episodic memory formation, offering insights into the temporal dynamics and causal interactions within these large-scale cognitive networks.

Our final objective was to investigate the replicability of our findings across multiple episodic memory domains involving both verbal and spatial materials. Reproducing findings across experiments is a significant challenge in neuroscience, particularly in invasive iEEG studies where data sharing and sample sizes have been notable limitations. There have been few previous replicated findings from human iEEG studies across multiple task domains. Quantitatively rigorous measures are needed to address the reproducibility crisis in human iEEG studies. We used Bayesian analysis to quantify the degree of replicability (Ly, Etz, Marsman, & Wagenmakers, 2019 [DOI](#); Verhagen & Wagenmakers, 2014 [DOI](#)). Bayes factors (BFs) are a powerful tool for evaluating evidence for replicability of findings across tasks and for determining the strength of evidence for the null hypothesis (Verhagen & Wagenmakers, 2014 [DOI](#)). Briefly, the replication BF is the ratio of marginal likelihood of the replication data, given the posterior distribution estimated from the original data, and the marginal likelihood for the replication data under the null hypothesis of no effect (Ly et al., 2019 [DOI](#)).

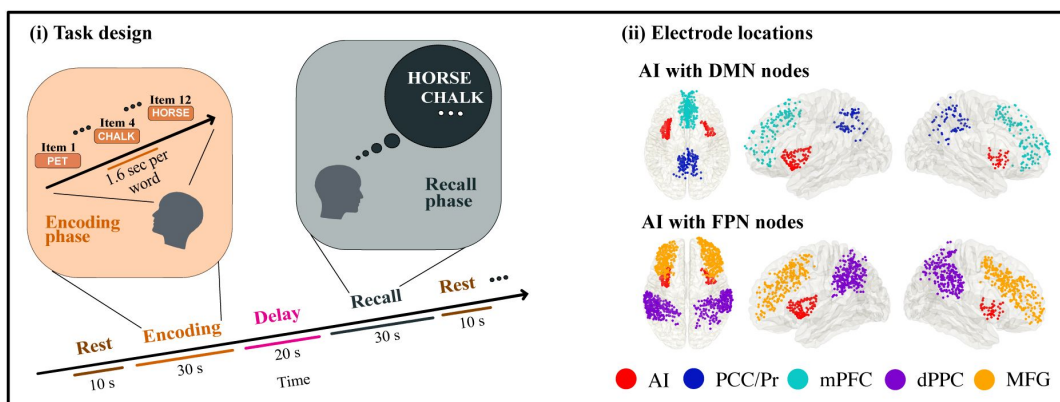
In summary, our study aims to elucidate the neurophysiological basis of the interactions between large-scale cognitive control networks by leveraging a unique dataset of iEEG recordings across multiple memory experiments. By examining directed causal information flow, high-gamma band power modulation, and replicability across verbal and spatial memory domains, we sought to advance our understanding of the neural mechanisms underpinning human episodic memory and cognitive control. Our findings shed light on how the brain effectively integrates information from distinct networks to support memory and cognition more broadly.

Results

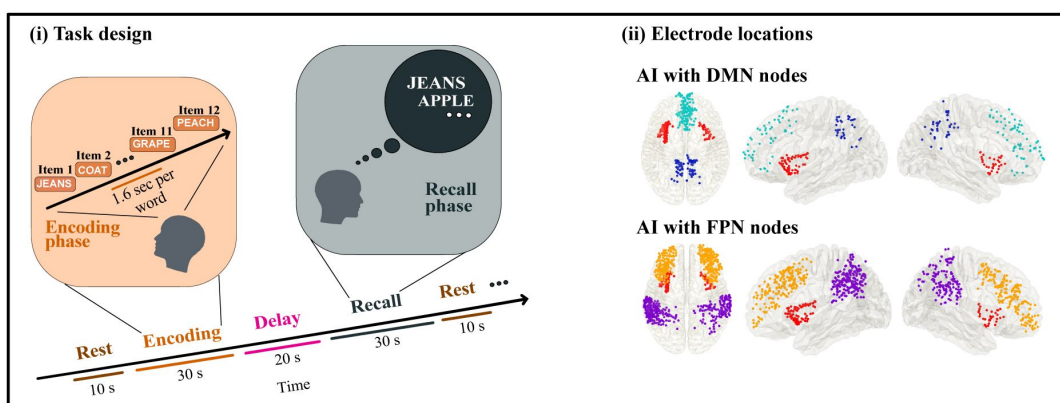
AI response compared to PCC/precuneus during encoding and recall in the VFR task

We first examined neuronal activity in the AI and the PCC/precuneus and tested whether activity in the PCC/precuneus is suppressed compared to activity in the AI. Previous studies have suggested that power in the high-gamma band (80-160 Hz) is correlated with fMRI BOLD signals (Hermes, Nguyen, & Winawer, 2017 [↗](#); Hutchison, Hashemi, Gati, Menon, & Everling, 2015 [↗](#); Lakatos, Gross, & Thut, 2019 [↗](#); Leopold, Murayama, & Logothetis, 2003 [↗](#); Mantini, Perrucci, Del Gratta, Romani, & Corbetta, 2007 [↗](#); Schölvinck, Maier, Ye, Duyn, & Leopold, 2010 [↗](#)), and is thought to reflect local neuronal activity (Canolty & Knight, 2010 [↗](#)). Therefore, we compared high-gamma band power (see **Methods** for details) in the AI and PCC/precuneus electrodes during both encoding and recall and across the four episodic memory tasks. Briefly, in the VFR task, participants were presented with a sequence of words and asked to remember them for subsequent recall (**Methods**, **Tables S1** [↗](#), **S2a** [↗](#), **S3a** [↗](#), **Figures 1a** [↗](#), **2** [↗](#)).

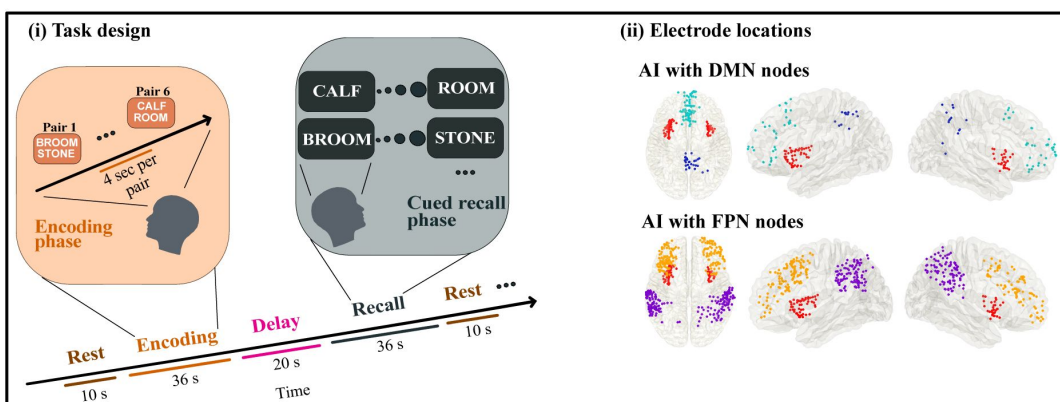
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

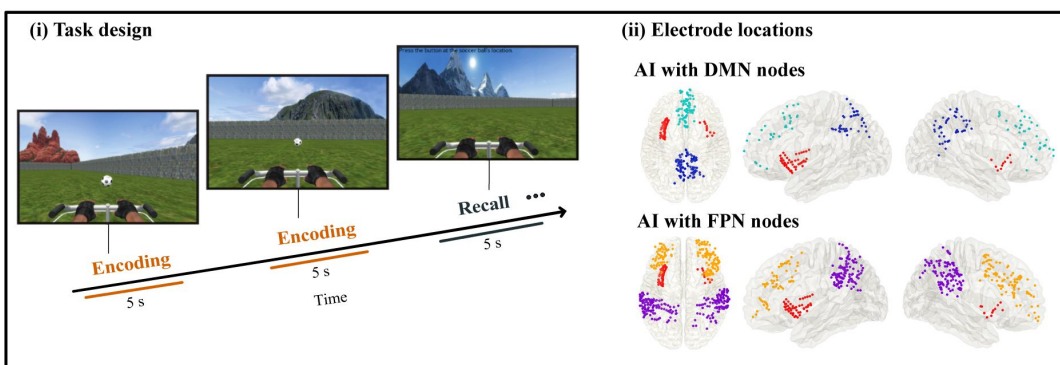


Figure 1.

Task design of the encoding and recall periods of the memory experiments, and iEEG recording sites in AI, with DMN and FPN nodes.

(a) Experiment 1, Verbal free recall (VFR): (i) Task design of memory encoding and recall periods of the verbal free recall experiment (see **Methods** for details). Participants were first presented with a list of words in the encoding block and asked to recall as many as possible from the original list after a short delay. (ii) Electrode locations for AI with DMN nodes (top panel) and AI with FPN nodes (bottom panel), in the verbal free recall experiment. Proportion of electrodes for AI, PCC/Pr, mPFC, dPPC, and MFG were 9%, 8%, 19%, 32%, and 32% respectively, in the VFR experiment. **(b) Experiment 2, Categorized verbal free recall (CATVFR):** (i) Task design of memory encoding and recall periods of the categorized verbal free recall experiment (see **Methods** for details). Participants were presented with a list of words with consecutive pairs of words from a specific category (for example, JEANS-COAT, GRAPE-PEACH, etc.) in the encoding block and subsequently asked to recall as many as possible from the original list after a short delay. (ii) Electrode locations for AI with DMN nodes (top panel) and AI with FPN nodes (bottom panel), in the categorized verbal free recall experiment. Proportion of electrodes for AI, PCC/Pr, mPFC, dPPC, and MFG were 10%, 7%, 11%, 35%, and 37% respectively, in the CATVFR experiment. **(c) Experiment 3, Paired associates learning verbal cued recall (PALVCR):** (i) Task design of memory encoding and recall periods of the paired associates learning verbal cued recall experiment (see **Methods** for details). Participants were first presented with a list of 6 word-pairs in the encoding block and after a short post-encoding delay, participants were shown a specific word-cue and asked to verbally recall the cued word from memory. (ii) Electrode locations for AI with DMN nodes (top panel) and AI with FPN nodes (bottom panel), in the paired associates learning verbal cued recall experiment. Proportion of electrodes for AI, PCC/Pr, mPFC, dPPC, and MFG were 14%, 5%, 13%, 33%, and 35% respectively, in the PALVCR experiment. **(d) Experiment 4, Water maze spatial memory (WMSM):** (i) Task design of memory encoding and recall periods of the water maze spatial memory experiment (see **Methods** for details). Participants were shown objects in various locations during the encoding period and asked to retrieve the location of the objects during the recall period. Modified from Jacobs et al. (2018) with permission. (ii) Electrode locations for AI with DMN nodes (top panel) and AI with FPN nodes (bottom panel), in the water maze spatial memory experiment. Proportion of electrodes for AI, PCC/Pr, mPFC, dPPC, and MFG were 10%, 15%, 13%, 38%, and 24% respectively, in the WMSM experiment. Overall, proportion of electrodes for VFR, CATVFR, PALVCR, and WMSM experiments were 43%, 27%, 15%, and 15% respectively. AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

Encoding

Compared to the AI, high-gamma power in PCC/precuneus was suppressed during almost the entire window 110 – 1600 msec during memory encoding ($p < 0.05$, **Figure 3a** [↗](#)).

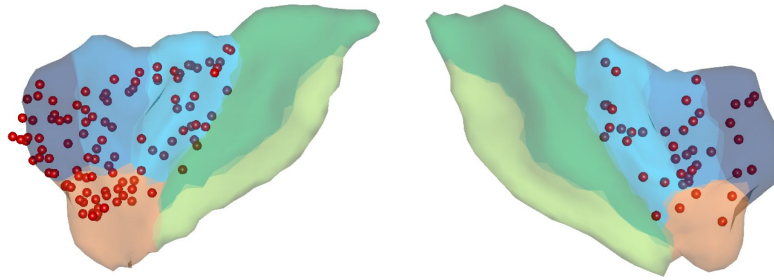
Recall

In contrast, suppression of high-gamma power in the PCC/precuneus was absent during the recall periods. Rather, high-gamma power in the PCC/precuneus was enhanced compared to the AI mostly during the 1390 – 1530 msec window prior to recall ($p < 0.05$, **Figure 3a** [↗](#)).

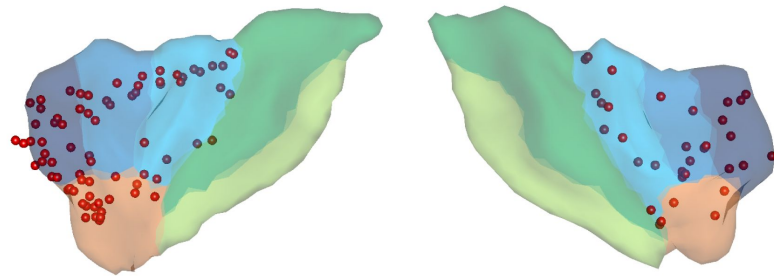
AI response compared to PCC/precuneus during encoding and recall in the CATVFR task

We next examined high-gamma power in the CATVFR task. In this task, participants were presented with a list of words with consecutive pairs of words from a specific category (for example, JEANS-COAT, GRAPE-PEACH, etc.) and subsequently asked to recall as many as possible from the original list (**Methods**, **Tables S1** [↗](#), **S2b** [↗](#), **S3b** [↗](#), **Figure 1b** [↗](#)) (Qasim, Mohan, Stein, & Jacobs, 2023 [↗](#)).

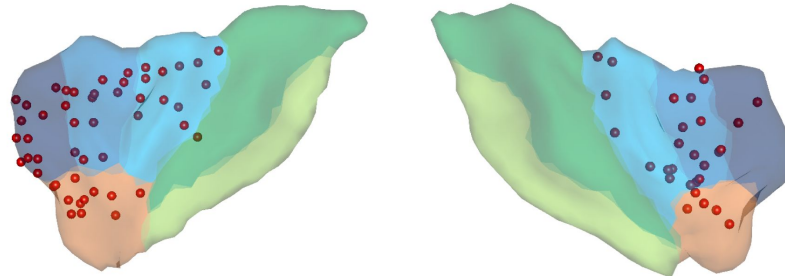
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

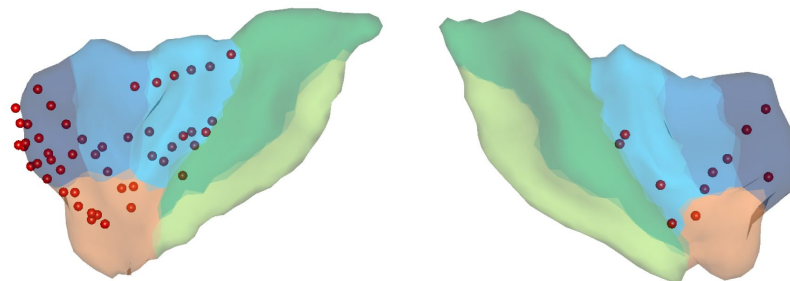
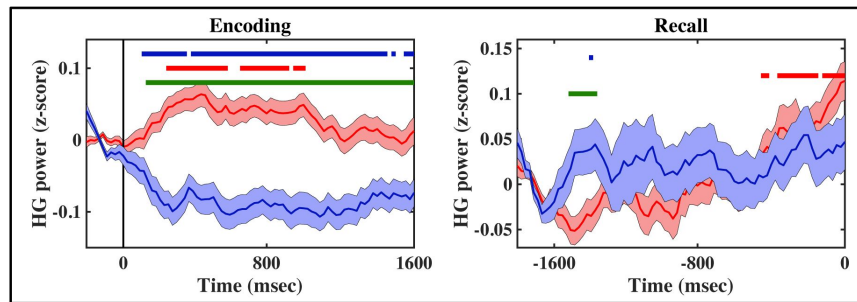


Figure 2.

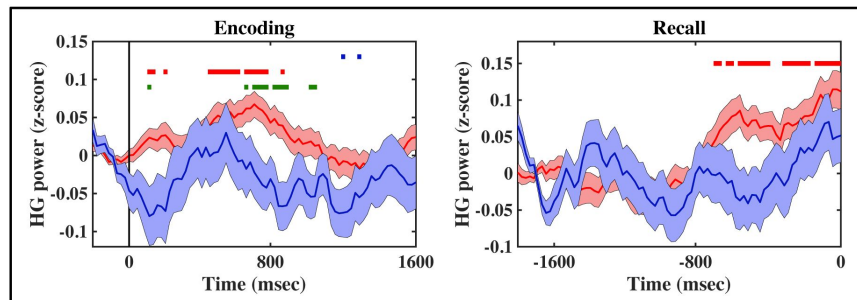
Anterior insula electrode locations (red) visualized on insular regions based on the atlas by Faillenot and colleagues (Faillenot, Heckemann, Frot, & Hammers, 2017 [\[1\]](#)).

Anterior insula is shown in blue, and posterior insula mask is shown in green (see **Methods** for details). This atlas is based on probabilistic analysis of the anatomy of the insula with demarcations of the AI based on three short dorsal gyri and the PI which encompasses two long and ventral gyri.

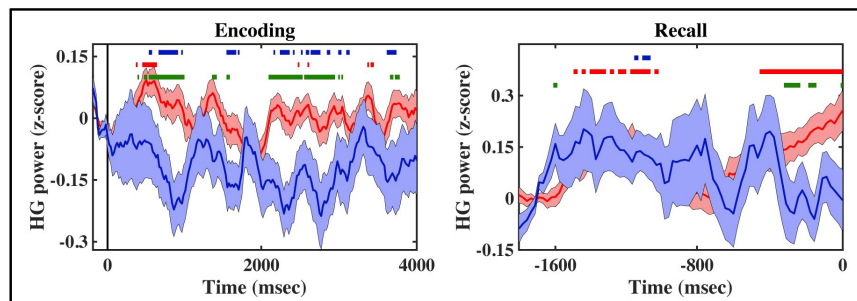
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

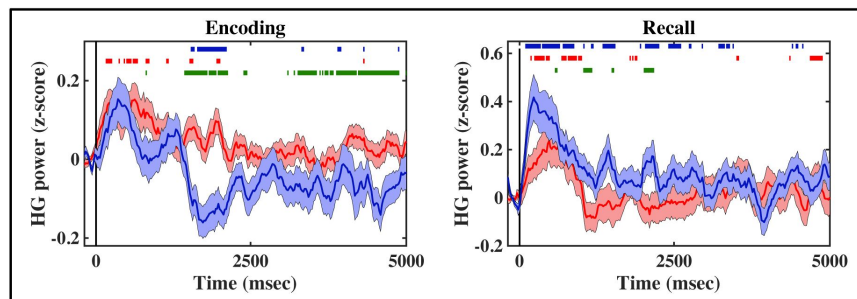


Figure 3.

iEEG evoked response, quantified using high-gamma (HG) power, for AI (red) and PCC/precuneus (blue) during (a) VFR, (b) CATVFR, (c) PALVCR, and (d) WMSM experiments.

Green horizontal lines denote high-gamma power for AI compared to PCC/precuneus ($p < 0.05$). Red horizontal lines denote increase of AI response compared to the resting baseline during the encoding and recall periods ($p < 0.05$). Blue horizontal lines denote decrease of PCC/precuneus response compared to the baseline during the encoding periods and increase of PCC/precuneus response compared to the baseline during the recall periods ($p < 0.05$).

Encoding

High-gamma power in PCC/precuneus was suppressed compared to the AI during the 570 – 790 msec interval ($p < 0.05$, [Figure 3b](#)).

Recall

High-gamma power mostly did not differ between AI and PCC/precuneus prior to recall ($p > 0.05$, [Figure 3b](#)).

AI response compared to PCC/precuneus during encoding and recall in the PALVCR task

The PALVCR task also consisted of three periods: encoding, delay, and recall (**Methods, Tables S1**, [S2c](#), [S3c](#), [Figure 1c](#)). During encoding, a list of word-pairs was visually presented, and then participants were asked to verbally recall the cued word from memory during the recall periods.

Encoding

High-gamma power in PCC/precuneus was suppressed compared to the AI during the memory encoding period, during the 470 – 950 msec and 2010 – 2790 msec windows ($p < 0.05$, [Figure 3c](#)).

Recall

High-gamma power mostly did not differ between AI and PCC/precuneus prior to recall ($p > 0.05$, [Figure 3c](#)).

AI response compared to PCC/precuneus during encoding and recall in the WMSM task

We next examined high-gamma power in the WMSM task. Participants performed multiple trials of a spatial memory task in a virtual navigation paradigm ([Goyal et al., 2018](#); [Jacobs et al., 2016](#); [Lee et al., 2018](#)) similar to the Morris water maze ([Morris, 1984](#)) (**Methods, Tables S1**, [S2d](#), [S3d](#), [Figure 1d](#)). Participants were shown objects in various locations during the encoding periods and asked to retrieve the location of the objects during the recall period.

Encoding

High-gamma power in PCC/precuneus was suppressed compared to the AI, mostly during the 1390 – 2030 msec and 3150 – 4690 msec window ($p < 0.05$, [Figure 3d](#)).

Recall

High-gamma power mostly did not differ between AI and PCC/precuneus ($p > 0.05$, [Figure 3d](#)).

Replication of increased high-gamma power in AI compared to PCC/precuneus across four memory tasks

We next used replication BF analysis to estimate the degree of replicability of high-gamma power suppression of the PCC/precuneus compared to the AI during the memory encoding periods of the four tasks. We used the posterior distribution obtained from the VFR (primary) dataset as a prior distribution for the test of data from the CATVFR, PALVCR, and WMSM (replication) datasets ([Ly et al., 2019](#)) (see **Methods** for details). We used the encoding time-windows for which we most

consistently observed decrease of PCC/precuneus high-gamma power compared to the AI. These correspond to 110 – 1600 msec during the VFR task, 570 – 790 msec in the CATVFR task, 2010 – 2790 msec in the PALVCR task, and 3150 – 4690 msec in the WMSM task. We first averaged the high-gamma power across these strongest time-windows for each task and then used replication BF analysis to estimate the degree of replicability of high-gamma power suppression of the PCC/precuneus compared to the AI.

Findings corresponding to the high-gamma power suppression of the PCC/precuneus compared to AI were replicated in the PALVCR (BF 5.16e+1) and WMSM (BF 2.69e+8) tasks. These results demonstrate very high replicability of high-gamma power suppression of the PCC/precuneus compared to AI during memory encoding. The consistent suppression effect was localized only to the PCC/precuneus, but not to the mPFC node of the DMN or the dPPC and MFG nodes of the FPN (Figures S1 [↗](#)-S3 [↗](#)).

In contrast to memory encoding, a similar analysis of high-gamma power did not reveal a consistent pattern of increased high-gamma power in AI and suppression of the PCC/precuneus across the four tasks during memory recall (Figure 3 [↗](#)).

AI and PCC/precuneus response during encoding and recall compared to resting baseline

We examined whether AI and PCC/precuneus high-gamma power response during the encoding and recall periods are enhanced or suppressed when compared to the baseline periods. High-gamma power in the AI was increased compared to the resting baseline during both the encoding and recall periods, and across all four tasks ($p < 0.05$, Figure 3 [↗](#)). This suggests an enhanced role for the AI during both memory encoding and recall compared to resting baseline.

In contrast, high-gamma power in the PCC/precuneus was reduced compared to the resting baseline in three tasks – VFR, PALVCR, and WMSM – providing direct evidence for PCC/precuneus suppression during memory encoding (Figure 3 [↗](#)). We did not find any increased high-gamma power activity in the PCC/precuneus, compared to the baseline, during memory retrieval (Figure 3 [↗](#)). These results provide evidence for PCC/precuneus suppression compared to both the AI and resting baseline, during externally triggered stimuli during encoding.

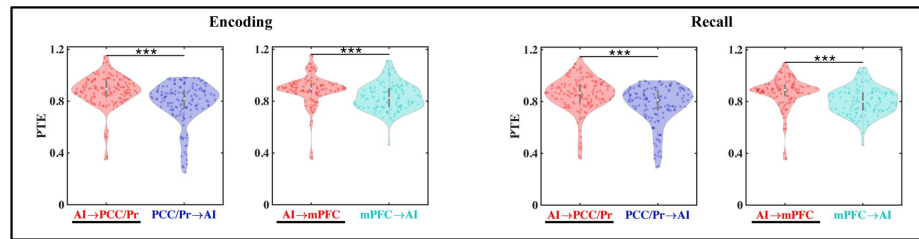
High-gamma power for other brain areas compared to resting baseline were not consistent across tasks (Figures S1 [↗](#)-S3 [↗](#)).

Causal information flow from the AI to the DMN during encoding

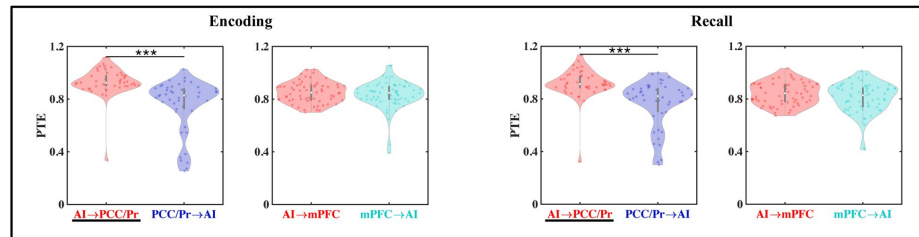
We next examined directed information flow from the AI to the PCC/precuneus and mPFC nodes of the DMN, during the memory encoding periods of the VFR task. We used phase transfer entropy (PTE) (Lobier et al., 2014 [↗](#)) to evaluate causal influences from the AI to the PCC/precuneus and mPFC and vice-versa. Informed by recent electrophysiology studies in nonhuman primates, which suggest that broadband field potentials activity, rather than narrowband, governs information flow in the brain (Davis, Muller, Martinez-Trujillo, Sejnowski, & Reynolds, 2020 [↗](#); Davis, Muller, & Reynolds, 2022 [↗](#)), we examined PTE in a 0.5 to 80 Hz frequency spectrum to assess dynamic causal influences of the AI on the DMN.

Directed information flow from the AI to the PCC/precuneus ($F(1, 264) = 59.36$, $p < 0.001$, Cohen's $d = 0.95$) and mPFC ($F(1, 208) = 13.96$, $p < 0.001$, Cohen's $d = 0.52$) were higher, than the reverse (Figure 4a [↗](#)).

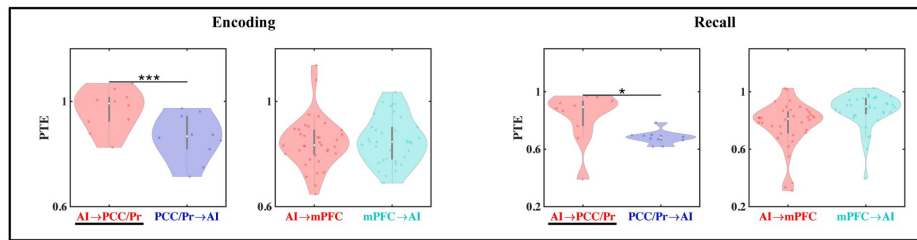
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

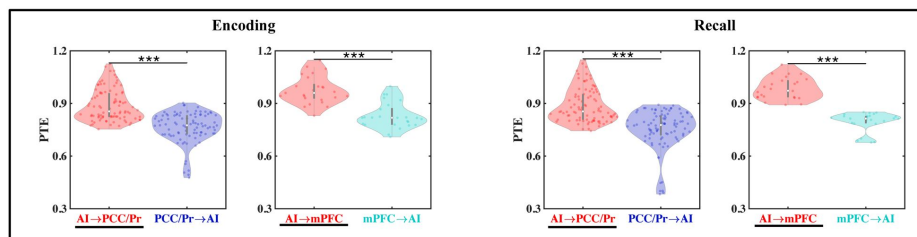


Figure 4.

Causal directed information flow between the anterior insula and the PCC/precuneus and mPFC nodes of the default mode network (DMN), across verbal and spatial memory domains, measured using phase transfer entropy (PTE).

(a) Experiment 1, VFR: The anterior insula showed higher causal directed information flow to the PCC/precuneus (AI → PCC/Pr) compared to the reverse direction (PCC/Pr → AI) ($n=142$) during both encoding and recall. The anterior insula also showed higher causal directed information flow to the mPFC (AI → mPFC) compared to the reverse direction (mPFC → AI) ($n=112$) during both memory encoding and recall. **(b) Experiment 2, CATVFR:** The anterior insula showed higher causal directed information flow to the PCC/precuneus (AI → PCC/Pr) compared to the reverse direction (PCC/Pr → AI) ($n=46$) during both encoding and recall. **(c) Experiment 3, PALVCR:** The anterior insula showed higher causal directed information flow to the PCC/precuneus (AI → PCC/Pr) compared to the reverse direction (PCC/Pr → AI) ($n=10$) during both encoding and recall. **(d) Experiment 4, WMSM:** The anterior insula showed higher causal directed information flow to PCC/precuneus (AI → PCC/Pr) than the reverse (PCC/Pr → AI) ($n=91$), during both spatial memory encoding and recall. The anterior insula also showed higher causal directed information flow to mPFC (AI → mPFC) than the reverse (mPFC → AI) ($n=23$), during both spatial memory encoding and recall. In each panel, the direction for which PTE is higher, is underlined. White dot in each violin plot represents median PTE across electrode pairs. *** $p < 0.001$, * $p < 0.05$.

Replication across three experiments with BF

We used replication BF analysis to estimate the degree of replicability of direction of information flow across the four experiments (**Table 1a** [↗](#), **Figures 4b-d** [↗](#), also see **Supplementary Results** for detailed stats related to the CATVFR, PALVCR, and WMSM experiments). Findings corresponding to the direction of information flow between the AI and the PCC/precuneus during memory encoding were replicated all three tasks (BFs $9.31\text{e}+5$, $1.44\text{e}+4$, and $1.68\text{e}+18$ for CATVFR, PALVCR, and WMSM respectively).

Findings corresponding to the direction of information flow between the AI and mPFC during memory encoding were also replicated across all three tasks (BFs $4.10\text{e}+1$, $8.78\text{e}+0$, and $5.34\text{e}+5$ for CATVFR, PALVCR, and WMSM respectively). This highly consistent pattern of results was not observed in any other frequency band (delta-theta (0.5-8 Hz), beta (12-30 Hz), gamma (30-80 Hz), and high-gamma (80-160 Hz); results not shown). These results demonstrate very high replicability of directed information flow from the AI to the DMN nodes during memory encoding.

These results demonstrate robust directed information flow from the AI to the PCC/precuneus and mPFC nodes of the DMN during memory encoding.

Causal information flow from the AI to the DMN during recall

Next, we examined causal influences of the AI on PCC/precuneus and mPFC during the recall phase of the verbal episodic memory task. During memory recall, directed information flow from the AI to the PCC/precuneus ($F(1, 264) = 43.09$, $p < 0.001$, Cohen's $d = 0.81$) and mPFC ($F(1, 211) = 21.94$, $p < 0.001$, Cohen's $d = 0.65$) were higher, than the reverse (**Figure 4a** [↗](#)).

Replication across three experiments with

BF We next repeated the replication BF analysis for the recall periods of the memory tasks (**Table 1b** [↗](#), **Figures 4b-d** [↗](#), also see **Supplementary Results** for detailed stats related to the CATVFR, PALVCR, and WMSM experiments). Findings corresponding to the direction of information flow between the AI and the PCC/precuneus during memory recall were replicated across all three tasks (BFs $1.30\text{e}+5$, $6.74\text{e}+0$, and $2.54\text{e}+10$ for CATVFR, PALVCR, and WMSM respectively). Findings corresponding to the direction of information flow between the AI and the mPFC during memory recall were also replicated across the CATVFR and WMSM tasks (BFs $2.02\text{e}+1$ and $1.32\text{e}+4$ respectively).

These results demonstrate very high replicability of directed information flow from the AI to the DMN nodes across verbal and spatial memory tasks, during both memory encoding and recall.

Causal information flow from AI to FPN nodes during memory encoding

We next probed directed information flow between the AI and FPN nodes during the encoding periods of the verbal free recall task. Directed information flow from the AI to the dPPC ($F(1, 1143) = 11.69$, $p < 0.001$, Cohen's $d = 0.20$) and MFG ($F(1, 1245) = 21.69$, $p < 0.001$, Cohen's $d = 0.26$) were higher, than the reverse during memory encoding of the VFR task (**Figure 5a** [↗](#)).

Replication across three experiments with BF

We used replication BF analysis for the replication of AI causal influences on FPN nodes during the encoding phase of the memory tasks (**Table 1a** [↗](#), **Figures 5b-d** [↗](#), **Supplementary Results**). Similarly, we also obtained very high BF for findings corresponding to the direction of

(a) Memory Encoding

Finding	Bayes Factor for VFR-CATVFR replication	Bayes Factor for VFR-PALVCR replication	Bayes Factor for VFR-WMSN replication
AI → PCC/Pr > PCC/Pr → AI	9.31e+5	1.44e+4	1.68e+18
AI → mPFC > mPFC → AI	4.10e+1	8.78e+0	5.34e+5
AI → dPPC > dPPC → AI	3.95e+43	2.33e+26	3.25e+40
AI → MFG > MFG → AI	1.49e+51	1.61e+33	2.35e+27

(b) Memory Recall

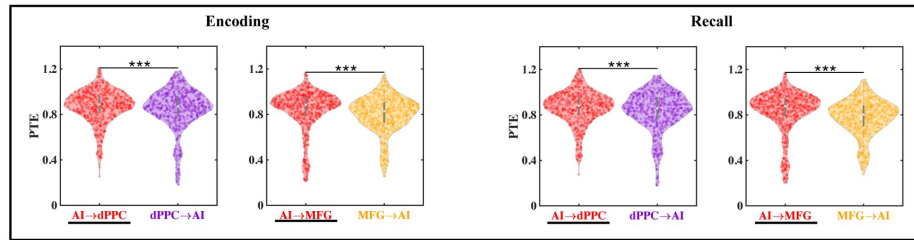
Finding	Bayes Factor for VFR-CATVFR replication	Bayes Factor for VFR-PALVCR replication	Bayes Factor for VFR-WMSN replication
AI → PCC/Pr > PCC/Pr → AI	1.30e+5	6.74e+0	2.54e+10
AI → mPFC > mPFC → AI	2.02e+1	3.52e-5	1.32e+4
AI → dPPC > dPPC → AI	7.04e+38	2.98e+45	4.51e+27
AI → MFG > MFG → AI	1.74e+54	5.72e+52	6.90e+27

Table 1.

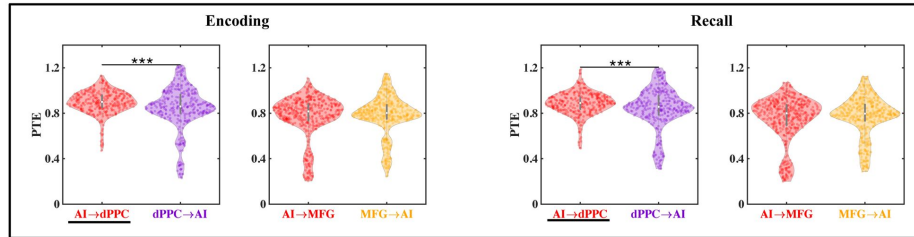
Replicability of findings of causal interactions of the AI with the DMN and FPN nodes for different memory experiments during (a) Memory Encoding and (b) Memory Recall.

The verbal free recall (VFR) task was considered the original dataset and the categorized verbal free recall (CATVFR), paired associates learning verbal cued recall (PALVCR), and water maze spatial memory (WMSM) tasks were considered replication datasets and Bayes factor (BF) for replication was calculated for pairwise tasks (verbal free recall vs. T, where T can be categorized verbal free recall, paired associates learning verbal cued recall, or water maze spatial memory task). Significant BF results (BF>3) are indicated in bold. AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

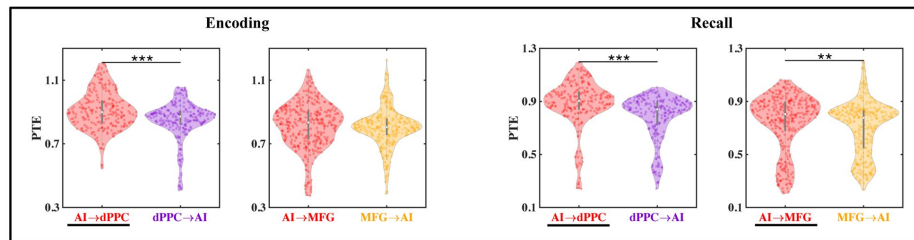
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

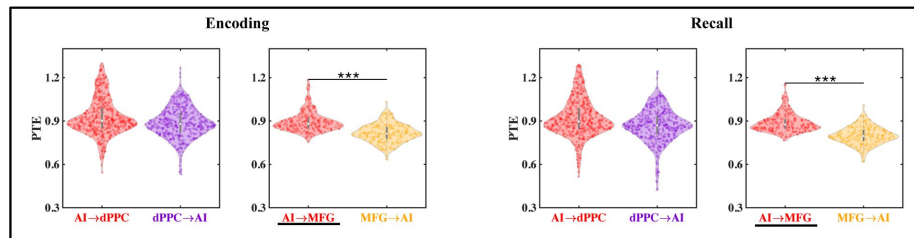


Figure 5.

Causal directed information flow between the anterior insula and the dPPC and MFG nodes of the frontoparietal network (FPN), across verbal and spatial memory domains.

(a) Experiment 1, VFR: The anterior insula showed higher causal directed information flow to the dorsal PPC (AI → dPPC) compared to the reverse direction (dPPC → AI) ($n=586$) during both encoding and recall. The anterior insula also showed higher causal directed information flow to the MFG (AI → MFG) compared to the reverse direction (MFG → AI) ($n=642$) during both memory encoding and recall. **(b) Experiment 2, CATVFR:** The anterior insula showed higher causal directed information flow to the dorsal PPC (AI → dPPC) compared to the reverse direction (dPPC → AI) ($n=327$) during both encoding and recall. **(c) Experiment 3, PALVCR:** The anterior insula showed higher causal directed information flow to the dorsal PPC (AI → dPPC) compared to the reverse direction (dPPC → AI) ($n=242$) during both encoding and recall. The anterior insula also showed higher causal directed information flow to the MFG (AI → MFG) compared to the reverse direction (MFG → AI) ($n=362$) during memory recall. **(d) Experiment 4, WMSM:** The anterior insula showed higher causal directed information flow to MFG (AI → MFG) than the reverse (MFG → AI) ($n=177$), during both spatial memory encoding and recall. In each panel, the direction for which PTE is higher, is underlined. *** $p < 0.001$, ** $p < 0.01$.

information flow between the AI and dPPC (BFs > 2.33e+26) and also between the AI and MFG (BFs > 2.35e+27), across all three tasks.

These results demonstrate that the AI has robust directed information flow to the dPPC and MFG nodes of the FPN during memory encoding.

Causal information flow from AI to FPN nodes during memory recall

Directed causal influences from the AI to the dPPC ($F(1, 1143) = 17.47, p < 0.001$, Cohen's $d = 0.25$) and MFG ($F(1, 1246) = 42.75, p < 0.001$, Cohen's $d = 0.37$) were higher, than the reverse during memory recall of the VFR task (Figure 5a [↗](#)).

Replication across three experiments with BF

We also found very high BFs for findings corresponding to the direction of information flow between the AI and the dPPC (BFs > 4.51e+27) and MFG (BFs > 6.90e+27) nodes of the FPN across the CATVFR, PALVCR, and WMSM tasks, during the memory recall period (Table 1b [↗](#), Figures 5b-d [↗](#), Supplementary Results).

These results demonstrate very high replicability of directed information flow from the AI to the FPN nodes across multiple memory experiments, during both memory encoding and recall.

Differential causal information flow from the AI to the DMN and FPN during episodic memory processing compared to resting baseline

We next examined whether differential causal directed information flow from the AI to the DMN and FPN nodes during the memory tasks differed from resting baseline. Resting baselines were extracted for each trial and the duration of task and rest epochs were matched to ensure that differences in network dynamics could not be explained by the differences in the duration of the epochs. We did not find any consistent directed causal influences from the AI on the DMN and FPN during memory encoding and recall, when compared to the resting baseline (Figures S4 [↗](#), S5 [↗](#)). These findings suggest that the AI has higher directed causal influence on both the DMN and FPN nodes than the reverse regardless of task-related increase or decrease of its causal influence compared to the resting baseline.

Differential causal information flow from the AI to the DMN and FPN for successfully recalled and forgotten memory trials

We examined memory effects by comparing PTE between successfully recalled and forgotten memory trials. However, this analysis did not reveal differences in causal influence from the AI on the DMN and FPN or the reverse, between successfully recalled and forgotten memory trials during the encoding as well as recall periods in any of the memory experiments (all $ps > 0.05$).

Causal outflow hub during encoding and recall

fMRI studies have suggested that the AI acts as a causal outflow hub with respect to interactions with the DMN and FPN (Sridharan et al., 2008 [↗](#)). To test the potential neural basis of this finding, we calculated net causal outflow as the difference between the total outgoing information and total incoming information (PTE(out)–PTE(in), see Methods for details).

Encoding

This analysis revealed that the net causal outflow from the AI is positive and higher than the PCC/precuneus ($F(1, 3319) = 154.8, p < 0.001$, Cohen's $d = 0.43$) node of the DMN, in the VFR task (**Figure 6a** [↗](#)).

This analysis also revealed that the net causal outflow from the AI is higher than both the dPPC ($F(1, 5346) = 67.87, p < 0.001$, Cohen's $d = 0.23$) and MFG ($F(1, 6920) = 132.74, p < 0.001$, Cohen's $d = 0.28$) nodes of the FPN, in the VFR task (**Figure 6a** [↗](#)).

Findings in the VFR task were also replicated across the CATVFR, PALVCR, and WMSM tasks, where we found that the net causal outflow from the AI is higher than the PCC/precuneus and mPFC nodes of the DMN and the dPPC and MFG nodes of the FPN (**Figures 6b-d** [↗](#), also see **Supplementary Results** for detailed stats related to the CATVFR, PALVCR, and WMSM experiments).

Recall

Net causal outflow from the AI is positive and higher than both PCC/precuneus ($F(1, 3287) = 151.21, p < 0.001$, Cohen's $d = 0.43$) and mPFC ($F(1, 4694) = 7.81, p < 0.01$, Cohen's $d = 0.08$) during the recall phase of the VFR task (**Figure 6a** [↗](#)).

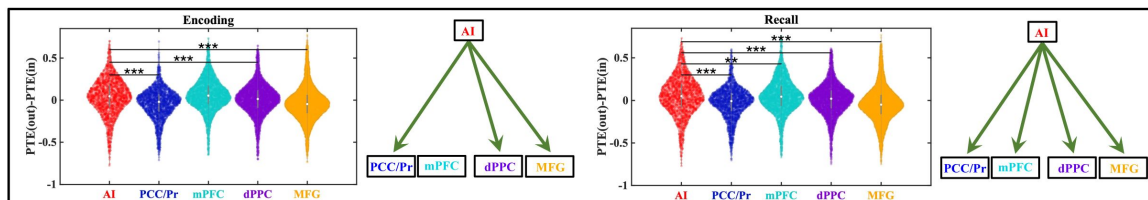
Net causal outflow from the AI is also higher than both the dPPC ($F(1, 5388) = 90.71, p < 0.001$, Cohen's $d = 0.26$) and MFG ($F(1, 6945) = 167.14, p < 0.001$, Cohen's $d = 0.31$) nodes of the FPN during recall (**Figure 6a** [↗](#)).

Crucially, these findings were also replicated across the CATVFR, PALVCR, and WMSM tasks and during both encoding and recall periods (**Figures 6b-d** [↗](#), also see **Supplementary Results** for detailed stats related to the CATVFR, PALVCR, and WMSM experiments). Together, these results demonstrate that the AI is a causal outflow hub in its interactions with the PCC/precuneus and mPFC nodes of the DMN and also the dPPC and MFG nodes of the FPN, during both verbal and spatial memory encoding and recall.

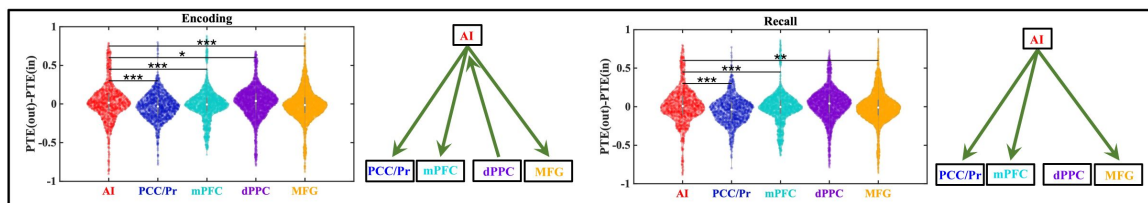
Discussion

Our study aimed to elucidate the electrophysiological basis of directed information flow within the triple network model, comprising key cortical nodes of the salience network (SN), default mode network (DMN), and frontoparietal network (FPN), during episodic memory formation. We discovered that the anterior insula (AI), a crucial node of the SN, exerts strong causal influence on both the DMN and FPN during memory encoding and recall. This finding was consistently observed across multiple experiments spanning verbal and spatial memory domains, highlighting the robustness and generalizability of our results. Importantly, our study extends the applicability of the triple network model beyond attention-demanding tasks to episodic memory processes, thus broadening its explanatory power in the context of cognitive control and memory formation. Furthermore, we observed a distinctive suppression of high-gamma power in the posterior cingulate cortex/precuneus (PCC/precuneus) node of the DMN compared to the AI during memory encoding, suggesting a task-specific functional down-regulation of this region. Our findings significantly advance the understanding of the SN's role in modulating large-scale brain networks during episodic memory formation and underscore the importance of the triple network model in elucidating the coordination of brain networks across various cognitive processes (**Figure 7** [↗](#)).

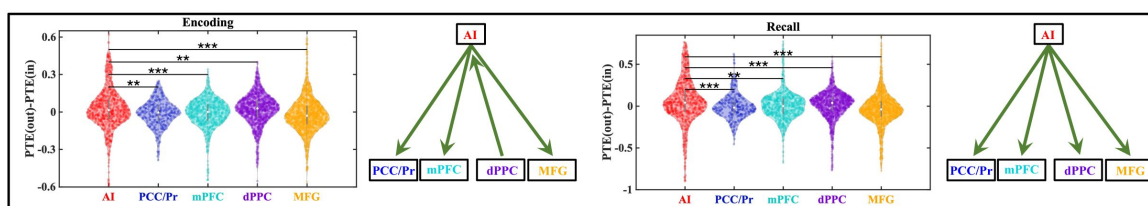
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

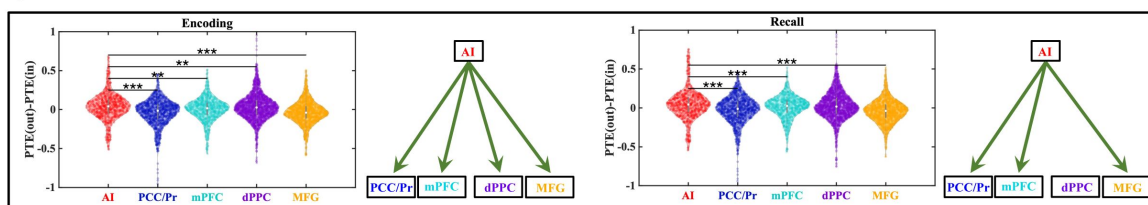


Figure 6.

The anterior insula is a causal outflow hub in its interactions with the DMN and FPN, during encoding and recall periods, and across memory experiments.

In each panel, the net direction of information flow between the AI and the DMN and FPN nodes are indicated by green arrows on the right. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

These results provide novel insights into the neurophysiological mechanisms underlying human cognition and memory, paving the way for a deeper understanding of the neural basis of cognitive deficits in various neuropsychological disorders.

Investigating causal inter-network interactions using iEEG

Dynamic causal interactions between the AI and the DMN and FPN are hypothesized to shape human cognition (Cai et al., 2016 [DOI](#); Cai, Ryali, Chen, Li, & Menon, 2014 [DOI](#); Dosenbach, Fair, Cohen, Schlaggar, & Petersen, 2008 [DOI](#); Dosenbach et al., 2006 [DOI](#); Menon, 2015 [DOI](#); Menon & Uddin, 2010 [DOI](#)). Although fMRI research has suggested that the AI plays a pivotal role in the task-dependent engagement and disengagement of the DMN and FPN across diverse cognitive tasks (Menon & Uddin, 2010 [DOI](#); Sridharan et al., 2008 [DOI](#)), the neuronal basis of these results or the possibility of their being artifacts arising from slow dynamics and regional variation in the hemodynamic response inherent to fMRI signals remained unclear. To address these ambiguities, our analysis focused on casual interactions involving the AI and leveraged the high temporal resolution of iEEG signals. By investigating the directionality of information flow, we aimed to overcome the temporal resolution limitations of fMRI signals, providing a more mechanistic understanding of the AI's role in modulating the DMN and FPN during memory formation. To assess reproducibility, we scrutinized network interactions across four different episodic memory tasks involving verbal free recall, categorized verbal free recall, paired associates learning verbal cued recall, and water maze spatial episodic memory tasks (Solomon et al., 2019 [DOI](#)).

We employed Phase Transfer Entropy (PTE), a robust metric of nonlinear and nonstationary causal dynamics to investigate dynamic causal interactions between the AI and four key cortical nodes of the DMN and FPN. PTE assesses the ability of one time-series to predict future values of another, estimating time-delayed causal influences, and is superior to methods like phase locking or coherence as it captures nonlinear and nonstationary interactions (Bassett & Sporns, 2017 [DOI](#); Hillebrand et al., 2016 [DOI](#); Lobier et al., 2014 [DOI](#)). PTE offers a robust and powerful tool for characterizing information flow between brain regions based on phase coupling (Hillebrand et al., 2016 [DOI](#); Lobier et al., 2014 [DOI](#); Wang et al., 2017 [DOI](#)) and has been successfully utilized in our previous studies (Das, de Los Angeles, & Menon, 2022 [DOI](#); Das & Menon, 2020 [DOI](#), 2021 [DOI](#), 2022b [DOI](#), 2023 [DOI](#)).

Broadband causal influences of the AI on DMN and FPN

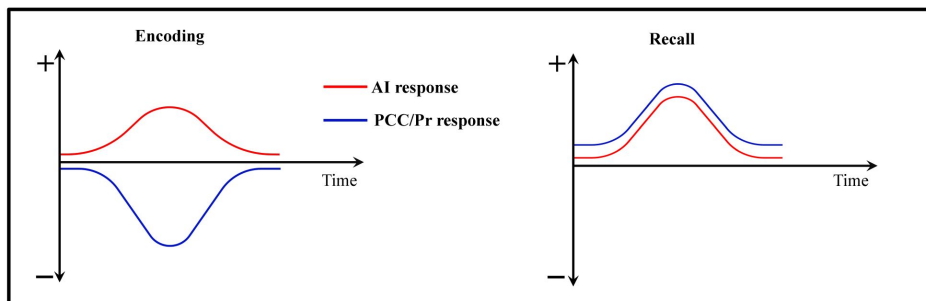
Informed by recent electrophysiology studies in nonhuman primates, which suggest that broadband field potentials activity, rather than narrowband, governs information flow in the brain (Davis et al., 2020 [DOI](#); Davis et al., 2022 [DOI](#)), we first examined PTE in a 0.5 to 80 Hz frequency spectrum to assess dynamic causal influences of the AI on the DMN and FPN. Our analysis revealed that AI exerts stronger causal influences on the PCC/precuneus and mPFC nodes of the DMN than the reverse. A similar pattern also emerged for FPN nodes, with the AI displaying stronger causal influences on the dPPC and MFG, than the reverse. Crucially, this asymmetric pattern of directed causal information flow was replicated across all four memory tasks. Moreover, this pattern also held during the encoding and recall of memory phases of all four tasks.

Replicability across memory tasks

Replication, a critical issue in all of systems neuroscience, is particularly challenging in the field of intracranial EEG studies, where data acquisition from patients is inherently difficult.

Compounding this issue is the virtual absence of data sharing and the substantial complexities involved in collecting electrophysiological data across distributed brain regions (Das & Menon, 2022b [DOI](#)). Consequently, one of our study's major objectives was to reproduce our findings across multiple experiments, bridging verbal and spatial memory domains and task phases. To quantify the degree of replicability of our findings across these domains, we employed replication Bayes Factor (BF) analysis (Ly et al., 2019 [DOI](#); Verhagen & Wagenmakers, 2014 [DOI](#)). Our analysis revealed

(a) High-gamma response



(b) Directed information flow

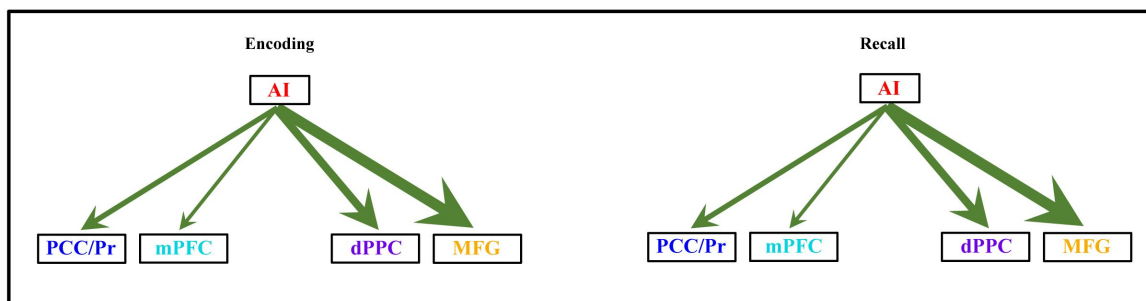


Figure 7.

Schematic illustration of key findings related to the intracranial electrophysiology of the triple network model in human episodic memory.

(a) High-gamma response: Our analysis of local neuronal activity revealed consistent suppression of high-gamma power in the PCC/precuneus compared to the AI during encoding periods across all four episodic memory experiments. We did not consistently observe any significant differences in high-gamma band power between AI and the mPFC node of the DMN or the dPPC and MFG nodes of the FPN during the encoding periods across the four episodic memory experiments. In contrast, we detected similar high-gamma band power in the PCC/precuneus relative to the AI during the recall periods. **(b) Directed information flow:** Despite variable patterns of local activation and suppression across DMN and FPN nodes during memory encoding and recall, we found stronger causal influence (denoted by green arrows, thickness of arrows denotes degree of replicability across experiments, see [Table 1](#)) by the AI on both the DMN as well as the FPN nodes than the reverse, across all four memory experiments, and during both encoding and recall periods.

very high replication BFs related to replication of causal information flow from the AI to the DMN and FPN (**Table 1**). Specifically, the BFs associated with the replication of direction of information flow between the AI and the DMN and FPN were decisive (BFs > 100), demonstrating consistent results across various memory tasks and contexts.

Externally triggered vs. internally driven memory processes

Our results reveal a consistent pattern of directed information flow between the AI and the DMN and FPN, which held true regardless of whether the tasks involved externally triggered stimuli during encoding and cued recall, or internally driven processes during free recall. This pattern underscores the robust and versatile role of the AI in modulating large-scale brain networks across diverse task contexts. In elucidating the interplay between externally and internally triggered memory processing, our study presents an intriguing finding. While we expected to observe a directed outflow from the AI during memory encoding triggered by external stimuli, the emergence of a similar pattern during internally triggered free recall was not initially foreseen. This reproducible pattern, observed across both externally and internally driven tasks, reinforces the crucial role of the AI in orchestrating network dynamics and advances our understanding of the intricate interplay between these large-scale networks in the human brain.

High-gamma power suppression in the PCC/precuneus during encoding, but not recall

The PCC/precuneus and mPFC, which form the central nodes of the DMN, are typically deactivated during attention demanding tasks (Wen et al., 2013). However, these regions also play an important direct role in episodic memory (Buckner et al., 2008; Menon, 2023). Recent iEEG studies have also shown a role of the PCC/precuneus for successful memory encoding (Natu et al., 2019; Tan, Rugg, & Lega, 2020). In contrast, the dPPC and MFG nodes of the FPN, which are integral to cognitive control over memory, typically display heightened activity during memory tasks (Badre et al., 2005; Badre & Wagner, 2007; Wagner et al., 2001; Wagner et al., 2005). Our analysis of local neuronal activity substantiated a differential activity pattern, revealing a consistent suppression of high-gamma power in the PCC/precuneus compared to the AI during encoding periods across all four episodic memory tasks. Intriguingly, within the DMN, this suppression effect was confined to the PCC/precuneus, with no parallel reductions observed in the mPFC. Furthermore, we did not observe any significant differences in high-gamma band power between AI and the dPPC and MFG nodes of the FPN. Replication analysis using Bayesian techniques substantiated a high degree of replicability in the suppression of high-gamma power in the PCC/precuneus vis-à-vis the AI during memory encoding (BFs > 5.16e+1).

These findings align with prior fMRI studies reporting DMN suppression during attention to external stimuli (Bressler & Menon, 2010; Raichle et al., 2001; Seeley et al., 2007) and are in line with research employing optogenetic stimulation of the AI in rodent brains that showcased dynamic DMN suppression patterns, particularly in the retrosplenial cortex (Menon et al., 2023). Our findings significantly extend this knowledge base to the specific domain of memory encoding, leveraging high-resolution iEEG recordings for temporal precision.

In contrast, our analysis of memory recall periods did not find a consistent pattern of increased high-gamma band power in AI and suppression of the PCC/precuneus across tasks. These findings underscore a consistent and specific pattern of suppression in the PCC/precuneus high-gamma power that is reliably present during the encoding periods of episodic memory tasks but absent during recall periods. The observed variance might stem from the differing cognitive demands—externally-stimulated effects during memory encoding versus internally-driven processes during free recall—characteristic of these two stages of memory.

High-gamma activity (typically ranging from 80-160 Hz) has been reliably implicated in various cognitive tasks across sensory modalities, including visual (Lachaux et al., 2005 [↗](#); Tallon-Baudry, Bertrand, Hénaff, Isnard, & Fischer, 2005 [↗](#)), auditory (Crone, Boatman, Gordon, & Hao, 2001 [↗](#); Edwards, Soltani, Deouell, Berger, & Knight, 2005 [↗](#)), and across cognitive domains, including working memory (Canolty et al., 2006 [↗](#); Mainy et al., 2007 [↗](#)) and episodic memory (Daitch & Parvizi, 2018 [↗](#); Sederberg et al., 2007 [↗](#)). This increase in high-gamma band activity is indicative of localized, task-related neural processing, often correlating with the synchronized activity of local neural populations (Canolty & Knight, 2010 [↗](#)). Specifically, increases in high-gamma power have been associated with elevated neuronal spiking and synaptic activity, rendering it a valuable marker of task-specific computations in local neuronal circuits (Ray, Crone, Niebur, Franaszczuk, & Hsiao, 2008 [↗](#)). In our study, we observed a distinctive pattern of suppression in PCC/precuneus high-gamma activity during memory encoding phases, but not during recall, compared to resting baseline. We suggest that a functional down-regulation is a plausible explanation of this high gamma suppression. This task-specific suppression indicates that the PCC/precuneus may have a specialized role in the regulation of attentional resources during the encoding phase of episodic memory formation.

Broadband vs. high-gamma causal influences

Notably, our findings reveal a robust and consistent causal influence exerted by the AI on all nodes of both the DMN and the FPN, extending across all four memory tasks and both memory encoding and recall phases. These causal influences were prominently manifested in broadband signals. Interestingly, such causal influences were not observed in the high-gamma frequency range (80-160 Hz). This absence aligns with current models positing that high-gamma activity is more likely to reflect localized processing, while lower-frequency bands are implicated in longer-range network communication and coordination (Bastos et al., 2015 [↗](#); Das et al., 2022 [↗](#); Das & Menon, 2020 [↗](#), 2021 [↗](#), 2022b [↗](#), 2023 [↗](#); K. J. Miller et al., 2007 [↗](#)). More generally, our findings emphasize that it is crucial to differentiate between high-gamma activity ($f > 80$ Hz) and sub-high-gamma ($f < 80$ Hz) fluctuations, as these signal types are indicative of different underlying physiological processes, each with distinct implications for understanding neural network dynamics.

Triple network model and memory processes

Our findings illuminate the interplay of large-scale brain networks in memory formation processes. The AI is considered a pivotal node in the triple network model of brain function, which encompasses the DMN and FPN, and the salience network. In this model, the AI, a key component of the salience network, is posited to act as a switch between the DMN and FPN. The investigation of activation (up-regulation) and deactivation (down-regulation) relative to the AI provides insights into the dynamics of how these large-scale networks interact during memory processes. More specifically, memory-related operations involve intricate interplays between different brain regions that are activated or deactivated depending on the stage of memory process— encoding or recall. By evaluating the causal influences of the AI, we were able to characterize the direction and intensity of information flow within these networks, thereby elucidating whether the triple network control processes are also applicable to memory tasks.

Insights beyond previous electrophysiology research

Previous electrophysiology research exploring intrinsic (resting-state) dynamics employed single-pulse electrical stimulation to probe the causal cortical dynamics associated with the salience network in parallel with electrode recordings in either the DMN or the FPN (Shine et al., 2017 [↗](#)). While the specific effect of AI stimulation was not examined, it was noted that stimulation of electrodes in the salience network, of which the AI is a major node, elicited a rapid (<70 ms) high-gamma band response, whereas stimulation of the DMN led to sustained responses in later time windows (85–200 ms). Delayed responses in high gamma band power have also been reported in

the DMN, compared to the AI, during a Go/NoGo task (Kucyi et al., 2020 [↗](#)). However, the direction and magnitude of information flow between the AI and the DMN was not directly analyzed. Our analysis of causal dynamics and direction of information flow involving the AI and the DMN and FPN fills a critical gap in the electrophysiology literature pertaining to network interactions during episodic memory. Importantly, given the frequency-specific effects of electrical stimulation, which can often exhibit opposing excitatory and inhibitory impacts (Grover, Nguyen, & Reinhart, 2021 [↗](#); Huang & Keller, 2022 [↗](#); Mercier et al., 2022 [↗](#)), causal influences associated with stimulation at a certain frequency may not necessarily mirror those occurring during task performance. Thus, our findings offer novel insights into the dynamic operation of cognitive control circuits during memory encoding and recall.

Successful and unsuccessful memory effects engage similar AI-directed causal circuits

Our analysis revealed no significant differences in directed connectivity between successfully recalled and forgotten memory trials, suggesting that the reported effects may not be specific to successful memory formation and may be related to attentional or other general cognitive processing rather than memory processing per se. While our study provides valuable insights into the interactions between the AI and the DMN and FPN during cognitive tasks involving verbal and spatial information processing during memory tasks, it is crucial to acknowledge that these interactions may not be unique to memory processes. The AI's causal influence on the DMN and FPN could reflect a more general role in coordinating attentional resources and cognitive control, which are essential for various cognitive functions, including memory formation (Menon & Uddin, 2010 [↗](#); Uddin, 2015 [↗](#)). To disentangle the specific contributions of memory recall and attention, future studies should incorporate carefully designed control tasks that do not involve memory components. It is also important to note that successful memory recall likely involves the coordinated activity of multiple brain systems beyond the triple network model investigated here. For instance, the medial temporal lobe, including the hippocampus and adjacent cortical regions, plays a crucial role in episodic memory formation and retrieval (Burgess, Maguire, & O'Keefe, 2002 [↗](#); Moscovitch et al., 2016 [↗](#)). Future studies will need to investigate a broader set of brain areas during successful and unsuccessful memory trials to gain a more comprehensive understanding of the neural circuits supporting distinctions between successfully recalled and forgotten memory trials.

AI as a causal outflow hub and a novel perspective on theoretical models of memory

Beyond information flow along individual pathways linking the AI with the DMN and FPN, our PTE analysis further revealed that the AI is a causal outflow hub in its interactions with the DMN and the FPN regardless of stimulus materials. As a central node of the salience network (Menon & Uddin, 2010 [↗](#); Seeley et al., 2007 [↗](#); Sridharan et al., 2008 [↗](#)), the AI is known to play a crucial role in influencing other networks (Menon & Uddin, 2010 [↗](#); Uddin, 2015 [↗](#)). Our results align with findings based on control theory analysis of brain networks during a working memory task. Specifically, Cai et al found higher causal outflow and controllability associated with the AI compared to DMN and FPN nodes during an n-back working memory task (Cai et al., 2021 [↗](#)). Controllability refers to the ability to perturb a system from a given initial state to other configuration states in finite time by means of external control inputs. Intuitively, nodes with higher controllability require lower energy for perturbing a system from a given state, making controllability measures useful for identifying driver nodes with the potential to influence overall state dynamics. By virtue of its higher controllability relative to other brain areas, the AI is well-positioned to dynamically engage and disengage with other brain areas. These findings expand our understanding of the AI's role, extending beyond attention and working memory tasks to incorporate two distinct stages of episodic memory formation. Our study, leveraging the temporal

precision of iEEG data, substantially enhances previous fMRI findings by unveiling the neurophysiological mechanisms underlying the AI's dynamic regulation of network activity during memory formation and cognition more generally.

Our findings bring a novel perspective to the seminal model of human memory proposed by Atkinson and Shiffrin (Atkinson & Shiffrin, 1968 [↗](#)). This model conceptualizes memory as a multistage process, with control mechanisms regulating the transition of information across these stages. The observed suppression of high-gamma power in the PCC/precuneus and enhancement in the AI during the encoding phase may be seen as one neurophysiological manifestation of these control processes. The AI's role as a dynamic switch, modulating activity between the DMN and FPN, aligns with active processing and control needed to encode sensory information into short-term memory. On the other hand, the transformations observed during the recall phase, particularly the discernable lack of DMN suppression patterns, may correspond to the retrieval processes where internally generated cues steer the reactivation of memory representations during recall. These results provide a novel neurophysiological model for understanding the complex control processes underpinning human memory functioning.

Limitations and future work

Our study has some constraints worth noting. Three out of the four memory experiments examined – VFR, CATVFR, and PALVCR – involved verbal recall which likely involves stronger left-lateralized activation compared to non-verbal memory encoding (Golby et al., 2001 [↗](#); Kelley et al., 1998 [↗](#); Nagel, Herting, Maxwell, Bruno, & Fair, 2013 [↗](#); Thomason et al., 2009 [↗](#)). Future studies with denser sampling of electrodes across both hemispheres and multiple verbal and non-verbal memory experiments are needed to separately examine the role of the individual hemispheres during verbal versus non-verbal memory retrieval. Additionally, our conclusions about the causal influence of the AI electrodes on the DMN and FPN electrodes were derived using computational methods. To more robustly establish these causal links, future studies will need to employ causal circuit manipulation techniques such as deep brain stimulation, using electrodes implanted in both the DMN and FPN while participants engage in episodic memory tasks. Finally, comparing the patterns of directed connectivity during memory and non-memory tasks may help elucidate the functional significance of these network interactions and their specific roles in memory and attentional processes. Despite these limitations, our findings lay the groundwork for future research aimed at delineating the electrophysiology of large-scale brain networks in supporting various cognitive functions, including episodic memory formation.

Conclusions

Our study provides novel insights into the neural dynamics underpinning episodic memory processing in humans by leveraging high temporal resolution iEEG recordings across a large cohort of participants engaged in four different episodic memory experiments. We discovered that the anterior insula (AI), a key node of the salience network, exerts a strong and consistent causal influence on both the default mode network (DMN) and frontoparietal network (FPN) during memory encoding and retrieval periods. This finding extends the applicability of the triple network model to episodic memory processes in both verbal and spatial domains, highlighting the AI's crucial role as a 'causal hub' that modulates information flow within and between these cognitive control networks during distinct stages of memory formation. Furthermore, our observation of a specific suppression of high-gamma power in the posterior cingulate cortex/precuneus (PCC/precuneus) node of the DMN during memory encoding – but not recall – suggests a task-specific functional down-regulation of this region, providing additional insights into operations of the triple network model. The robust reproducibility of our findings across multiple memory tasks underscores the reliability and generalizability of our results.

More broadly, our results highlight signaling across distributed large-scale brain networks during episodic memory processing, reinforcing the concept that memory operations are reliant on the concerted action of an ensemble of widely distributed brain networks (Mesulam, 1990 [↗](#)). By elucidating the electrophysiological basis of directed information flow within the triple network model, our study advances the understanding of neural circuit dynamics and their role in human cognition and memory.

Finally, our findings have potential clinical implications for understanding the neural basis of memory impairments in various neurological and psychiatric disorders. For instance, Alzheimer's disease is known to be intricately associated with dysfunctions in the salience, default mode, and frontoparietal networks (Bonthius, Solodkin, & Van Hoesen, 2005 [↗](#); Guzmán-Vélez et al., 2022 [↗](#)). Our study provides a template for future research aimed at developing targeted interventions and therapeutic strategies for cognitive disorders.

Conflict of interest statement

The authors declare no competing financial interests.

Acknowledgements

This research was supported by NIH grants NS086085 and MH126518. We are grateful to members of the UPENN-RAM consortia for generously sharing their unique iEEG data. We thank Dr. Byeongwook Lee for assistance with the figures. We acknowledge the computational resources and support provided by the Stanford Research Computing Center.

Methods

UPENN-RAM iEEG recordings

iEEG recordings from 249 patients shared by Kahana and colleagues at the University of Pennsylvania (UPENN) (obtained from the UPENN-RAM public data release) were used for analysis (Jacobs et al., 2016 [↗](#)). Patients with pharmaco-resistant epilepsy underwent surgery for removal of their seizure onset zones. iEEG recordings of these patients were downloaded from a UPENN-RAM consortium hosted data sharing archive (URL: <http://memory.psych.upenn.edu/RAM> [↗](#)). Prior to data collection, research protocols and ethical guidelines were approved by the Institutional Review Board at the participating hospitals and informed consent was obtained from the participants and guardians (Jacobs et al., 2016 [↗](#)).

Details of all the recording sessions and data pre-processing procedures are described by Kahana and colleagues (Jacobs et al., 2016 [↗](#)). Briefly, iEEG recordings were obtained using subdural grids and strips (contacts placed 10 mm apart) or depth electrodes (contacts spaced 5–10 mm apart) using recording systems at each clinical site. iEEG systems included DeltaMed XITek (Natus), Grass Telefactor, and Nihon-Kohden EEG systems. Electrodes located in brain lesions or those which corresponded to seizure onset zones or had significant interictal spiking or had broken leads, were excluded from analysis.

Anatomical localization of electrode placement was accomplished by co-registering the postoperative computed CTs with the postoperative MRIs using FSL (FMRIB (Functional MRI of the Brain) Software Library), BET (Brain Extraction Tool), and FLIRT (FMRIB Linear Image

Registration Tool) software packages. Preoperative MRIs were used when postoperative MRIs were not available. The resulting contact locations were mapped to MNI space using an indirect stereotactic technique and OsiriX Imaging Software DICOM viewer package.

We used the insula atlas by Faillenot and colleagues to demarcate the anterior insula (AI) (Faillenot et al., 2017 [DOI](#)), downloaded from <http://brain-development.org/brain-atlases/adult-brain-atlases/>. This atlas is based on probabilistic analysis of the anatomy of the insula with demarcations of the AI based on three short dorsal gyri and the posterior insula (PI) which encompasses two long gyri. To visualize iEEG electrodes on the insula atlas, we used surface-rendering code (GitHub: <https://github.com/ludovicbellier/InsulaWM>) provided by Llorens and colleagues (Llorens et al., 2023 [DOI](#)). We used the Brainnetome atlas (Fan et al., 2016 [DOI](#)) to demarcate the posterior cingulate cortex (PCC)/precuneus, the medial prefrontal cortex (mPFC), the dorsal posterior parietal cortex (dPPC), and the middle frontal gyrus (MFG). The dorsal anterior cingulate cortex node of the salience network was excluded from analysis due to lack of sufficient electrode placement. Out of 249 individuals, data from 177 individuals (aged from 16 to 64, mean age 36.3 ± 11.5 , 91 females) were used for subsequent analysis based on electrode placement in the AI and the PCC/precuneus, mPFC, dPPC, and MFG.

Original sampling rates of iEEG signals were 500 Hz, 1000 Hz, 1024 Hz, and 1600 Hz. Hence, iEEG signals were downsampled to 500 Hz, if the original sampling rate was higher, for all subsequent analysis. The two major concerns when analyzing interactions between closely spaced intracranial electrodes are volume conduction and confounding interactions with the reference electrode (Burke et al., 2013 [DOI](#); Frauscher et al., 2018 [DOI](#)). Hence bipolar referencing was used to eliminate confounding artifacts and improve the signal-to-noise ratio of the neural signals, consistent with previous studies using UPENN-RAM iEEG data (Burke et al., 2013 [DOI](#); Ezzyat et al., 2018 [DOI](#)). Signals recorded at individual electrodes were converted to a bipolar montage by computing the difference in signal between adjacent electrode pairs on each strip, grid, and depth electrode and the resulting bipolar signals were treated as new “virtual” electrodes originating from the midpoint between each contact pair, identical to procedures in previous studies using UPENN-RAM data (Solomon et al., 2019 [DOI](#)). Line noise (60 Hz) and its harmonics were removed from the bipolar signals and finally each bipolar signal was Z-normalized by removing mean and scaling by the standard deviation. For filtering, we used a fourth order two-way zero phase lag Butterworth filter throughout the analysis. iEEG signals were filtered in the broad frequency spectrum (0.5-80 Hz) as well as narrowband frequency spectra delta-theta (0.5-8 Hz), beta (12-30 Hz), gamma (30-80 Hz), and high-gamma (80-160 Hz).

Episodic memory experiments

(a) Verbal free recall (VFR) task

Patients performed multiple trials of a verbal free recall experiment, where they were presented with a list of words and subsequently asked to recall as many as possible from the original list (Figure 1a [DOI](#)) (Solomon et al., 2017 [DOI](#); Solomon et al., 2019 [DOI](#)). The task consisted of three periods: encoding, delay, and recall. During encoding, a list of 12 words was visually presented for ~30 sec. Words were selected at random, without replacement, from a pool of high frequency English nouns (http://memory.psych.upenn.edu/Word_Pools). Each word was presented for a duration of 1600 msec, followed by an inter-stimulus interval of 800 to 1200 msec. After the encoding period, participants engaged in a math distractor task (the delay period in Figure 1a [DOI](#)), where they were instructed to solve a series of arithmetic problems in the form of $a + b + c = ??$, where a , b , and c were randomly selected integers ranging from 1 to 9. Mean accuracy across patients in the math task was $90.87\% \pm 7.22\%$, indicating that participants performed the math task with a high level of accuracy, similar to our previous studies (Das & Menon, 2022a [DOI](#)). After a 20 sec post-encoding delay, participants were instructed to recall as many words as possible during the 30 sec recall period. Average recall accuracy across patients was $25.0\% \pm 10.6\%$, similar to prior studies of

verbal episodic memory retrieval in neurosurgical patients (Burke et al., 2014 [↗](#)). We analyzed iEEG epochs from the encoding and recall periods of the verbal free recall task. For the recall periods, iEEG recordings 1600 msec prior to the vocal onset of each word were analyzed (Solomon et al., 2019 [↗](#)). Data from each trial was analyzed separately and specific measures were averaged across trials.

(b) Categorized verbal free recall (CATVFR) task

This task was very similar to the verbal free recall task. Here, patients performed multiple trials of a categorized free recall experiment, where they were presented with a list of words with consecutive pairs of words from a specific category (for example, JEANS-COAT, GRAPE-PEACH, etc.) and subsequently asked to recall as many as possible from the original list (**Figure 1b** [↗](#)) (Qasim et al., 2023 [↗](#)). Similar to the uncategorized verbal free recall task, this task also consisted of three periods: encoding, delay, and recall. During encoding, a list of 12 words was visually presented for ~30 sec. Semantic categories were chosen using Amazon Mechanical Turk. Pairs of words from the same semantic category were never presented consecutively. Each word was presented for a duration of 1600 msec, followed by an inter-stimulus interval of 750 to 1000 msec. After a 20 sec post-encoding delay (math) similar to the uncategorized verbal free recall task, participants were instructed to recall as many words as possible during the 30 sec recall period. Average accuracy across patients in the math task was $89.46\% \pm 9.90\%$. Average recall accuracy across patients was $29.6\% \pm 13.4\%$. Analysis of iEEG epochs from the encoding and recall periods of the categorized free recall task was same as the uncategorized verbal free recall task.

(c) Paired associates learning verbal cued recall (PALVCR) task

Patients performed multiple trials of a paired associates learning verbal cued recall experiment, where they were presented with a list of word-pairs and subsequently asked to recall based on the given word-cue (**Figure 1c** [↗](#)). Similar to the uncategorized verbal free recall task, this task also consisted of three periods: encoding, delay, and recall. During encoding, a list of 6 word-pairs was visually presented for ~36 sec. Similar to the uncategorized verbal free recall task, words were selected at random, without replacement, from a pool of high frequency English nouns (http://memory.psych.upenn.edu/Word_Pools [↗](#)). Each word was presented for a duration of 4000 msec, followed by an inter-stimulus interval of 1750 to 2000 msec. After a 20 sec post-encoding delay (math) similar to the uncategorized verbal free recall task, participants were shown a specific word-cue for a duration of 4000 msec and asked to verbally recall the cued word from memory. Each word presentation during recall was followed by an inter-stimulus interval of 1750 to 2000 msec and the recall period lasted for ~36 sec. Average accuracy across patients in the math task was $93.91\% \pm 4.66\%$. Average recall accuracy across patients was $33.8\% \pm 25.9\%$. For encoding, iEEG recordings corresponding to the 4000 msec encoding period of the task were analyzed. For recall, iEEG recordings 1600 msec prior to the vocal onset of each word were analyzed (Solomon et al., 2019 [↗](#)). Data from each trial was analyzed separately and specific measures were averaged across trials.

(d) Water maze spatial memory (WMSM) task

Patients performed multiple trials of a spatial memory experiment in a virtual navigation paradigm (Goyal et al., 2018 [↗](#); Jacobs et al., 2016 [↗](#); Lee et al., 2018 [↗](#)) similar to the Morris water maze (Morris, 1984 [↗](#)). The environment was rectangular (1.8:1 aspect ratio), and was surrounded by a continuous boundary (**Figure 1d** [↗](#)). There were four distal visual cues (landmarks), one centered on each side of the rectangle, to aid with orienting. Each trial (96 trials per session, 1–3 sessions per subject) started with two 5 sec encoding periods, during which subjects were driven to an object from a random starting location. At the beginning of an encoding period, the object appeared and, over the course of 5 sec, the subject was automatically driven directly toward it. The 5 sec period consisted of three intervals: first, the subject was rotated toward the object (1 sec), second, the subject was driven toward the object (3 sec), and, finally, the subject paused while

at the object location (1 sec). After a 5 sec delay with a blank screen, the same process was repeated from a different starting location. After both encoding periods for each item, there was a 5 sec pause followed by the recall period. The subject was placed in the environment at a random starting location with the object hidden and then asked to freely navigate using a joystick to the location where they thought the object was located. When they reached their chosen location, they pressed a button to record their response. They then received feedback on their performance via an overhead view of the environment showing the actual and reported object locations. Average recall accuracy across patients was $48.1\% \pm 5.6\%$.

We analyzed the 5 sec iEEG epochs corresponding to the entire encoding and recall periods of the task as has been done previously (Goyal et al., 2018 [DOI](#); Jacobs et al., 2016 [DOI](#); Lee et al., 2018 [DOI](#)). Data from each trial was analyzed separately and specific measures were averaged across trials, similar to the verbal tasks.

Out of total 177 participants, 51% (91 out of 177) of participants participated in at least 2 experiments, 17% (30 out of 177) of participants participated in at least 3 experiments, and 6% (10 out of 177) of participants participated in all four experiments.

iEEG analysis of high-gamma power

We first filtered the signals in the high-gamma (80-160 Hz) frequency band (Canolty et al., 2006 [DOI](#); Helfrich & Knight, 2016 [DOI](#); Kai J. Miller, Weaver, & Ojemann, 2009 [DOI](#)) and then calculated the square of the filtered signals as the power of the signals (Kwon et al., 2021 [DOI](#)). Signals were then smoothed using 0.2s windows with 90% overlap (Kwon et al., 2021 [DOI](#)) and normalized with respect to 0.2s pre-stimulus periods.

iEEG analysis of phase transfer entropy (PTE) and causal dynamics

Phase transfer entropy (PTE) is a nonlinear measure of the directionality of information flow between time-series and can be applied as a measure of causality to nonstationary time-series (Das & Menon, 2021 [DOI](#), 2022b [DOI](#); Lobier et al., 2014 [DOI](#)). Note that information flow described here relates to signaling between brain areas and does not necessarily reflect the representation or coding of behaviorally relevant variables per se. The PTE measure is in contrast to the Granger causality measure which can be applied only to stationary time-series (Barnett & Seth, 2014 [DOI](#)). We first carried out a stationarity test of the iEEG recordings (unit root test for stationarity (Barnett & Seth, 2014 [DOI](#))) and found that the spectral radius of the autoregressive model is very close to one, indicating that the iEEG time-series is nonstationary. This precluded the applicability of the Granger causality analysis in our study.

Given two time-series $\{x_i\}$ and $\{y_i\}$, where $i = 1, 2, \dots, M$, instantaneous phases were first extracted using the Hilbert transform. Let $\{x_i^p\}$ and $\{y_i^p\}$, where $i = 1, 2, \dots, M$, denote the corresponding phase time-series. If the uncertainty of the target signal $\{y_i^p\}$ at delay τ is quantified using Shannon entropy, then the PTE from driver signal $\{x_i^p\}$ to target signal $\{y_i^p\}$ can be given by

$$PTE_{x \rightarrow y} = \sum_i p(y_{i+\tau}^p, y_i^p, x_i^p) \log \left(\frac{p(y_{i+\tau}^p | y_i^p, x_i^p)}{p(y_{i+\tau}^p | y_i^p)} \right), \quad (i)$$

where the probabilities can be calculated by building histograms of occurrences of singles, pairs, or triplets of instantaneous phase estimates from the phase time-series (Hillebrand et al., 2016 [DOI](#)). For our analysis, the number of bins in the histograms was set as $3.49 \times STD \times M^{-1/3}$ and delay τ was set as $2M / M_{\pm}$, where STD is average standard deviation of the phase time-series $\{x_i^p\}$ and $\{y_i^p\}$ and $M_{\pm}$ is the number of times the phase changes sign across time and channels (Hillebrand et al.,

2016 [2016](#)). PTE has been shown to be robust against the choice of the delay τ and the number of bins for forming the histograms (Hillebrand et al., 2016 [2016](#)). In our analysis, PTE was calculated for the entire encoding and recall periods for each trial and then averaged across trials.

Net causal outflow was calculated as the difference between the total outgoing information and total incoming information, that is, net causal outflow = PTE(out) – PTE(in). For example, for calculation of PTE(out) and PTE(in) for the AI electrodes, electrodes in the PCC/precuneus, mPFC, dPPC, and MFG were considered, that is, PTE(out) was calculated as the net PTE from AI electrodes to the PCC/precuneus, mPFC, dPPC, and MFG electrodes, and PTE(in) was calculated as the net PTE from the PCC/precuneus, mPFC, dPPC, and MFG electrodes to AI electrodes. Net causal outflow for the PCC/precuneus, mPFC, dPPC, and MFG electrodes were calculated similarly.

Statistical analysis

Statistical analysis was conducted using mixed effects analysis with the lmerTest package (Kuznetsova, Brockhoff, & Christensen, 2017 [2017](#)) implemented in R software (version 4.0.2, R Foundation for Statistical Computing). Because PTE data were not normally distributed, we used BestNormalize (Peterson & Cavanaugh, 2018 [2018](#)) which contains a suite of transformation-estimating functions that can be used to optimally normalize data. The resulting normally distributed data were subjected to mixed effects analysis with the following model: $PTE \sim Condition + (1 | Subject)$, where *Condition* models the fixed effects (condition differences) and $(1 | Subject)$ models the random repeated measurements within the same participant, similar to prior iEEG studies (Das & Menon, 2021 [2021](#); Hoy, Steiner, & Knight, 2021 [2021](#); Salamone et al., 2021 [2021](#)). Before running the mixed-effects model, PTE was first averaged across trials for each channel pair. Analysis of variance (ANOVA) was used to test the significance of findings with FDR-corrections for multiple comparisons ($p < 0.05$). Linear mixed effects models were run for encoding and recall periods separately. Similar mixed effects statistical analysis procedures were used for comparison of high-gamma power across task conditions, where the mixed effects analysis was run on each of the 0.2s windows.

For effect size estimation, we used Cohen's *d* statistics for pairwise comparisons. We used the *lme.dscore()* function in the *EMAtools* package in R for estimating Cohen's *d*.

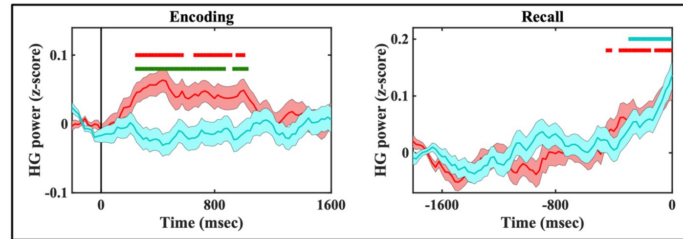
Bayesian replication analysis

We used replication Bayes factor (Ly et al., 2019 [2019](#); Verhagen & Wagenmakers, 2014 [2014](#)) analysis to estimate the degree of replicability for the direction of information flow for each frequency and task condition and across task domains. Analysis was implemented in R software using the BayesFactor package (Rouder, Speckman, Sun, Morey, & Iverson, 2009 [2009](#)). Because PTE data were not normally distributed, as previously, we used BestNormalize (Peterson & Cavanaugh, 2018 [2018](#)) to optimally normalize data. We calculated the replication Bayes factor for pairwise experiments. We compared the Bayes factor of the joint model $PTE(task1+task2) \sim Condition + (1 | Subject)$ with the Bayes factor (BF) of individual model as $PTE(task1) \sim Condition + (1 | Subject)$, where *task1* denotes the verbal free recall (original) task and *task2* denotes the categorized verbal free recall, paired associates learning verbal cued recall, or water maze spatial memory (replication) conditions. We calculated the ratio $BF(task1+task2)/BF(task1)$, which was used to quantify the degree of replicability. We determined whether the degree of replicability was higher than 3 as Bayes factor of at least 3 indicates evidence for replicability (Jeffreys, 1998 [1998](#)). A Bayes factor of at least 100 is considered as “decisive” for degree of replication (Jeffreys, 1998 [1998](#)). Same analysis procedures were used to estimate the degree of replicability for high-gamma power comparison of DMN and FPN electrodes with the AI electrodes, across experiments.

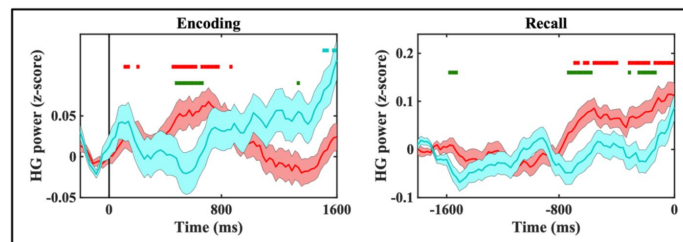
Supplementary Materials

Supplementary Figures

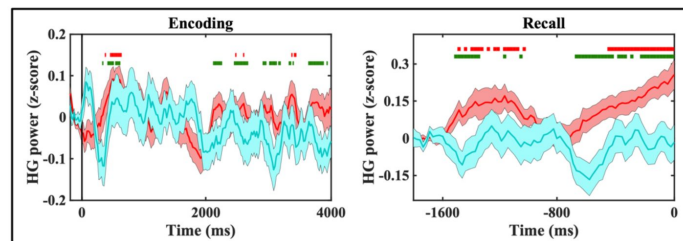
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

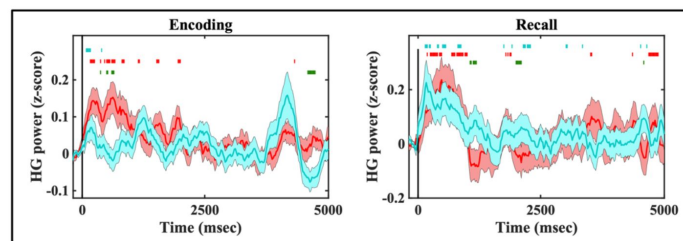
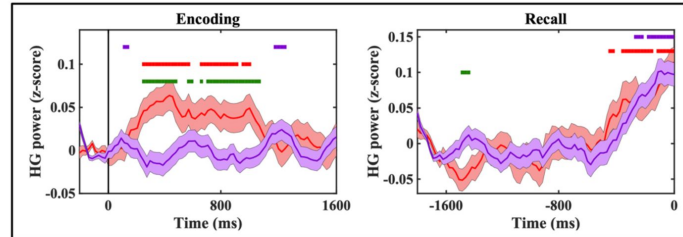


Figure S1.

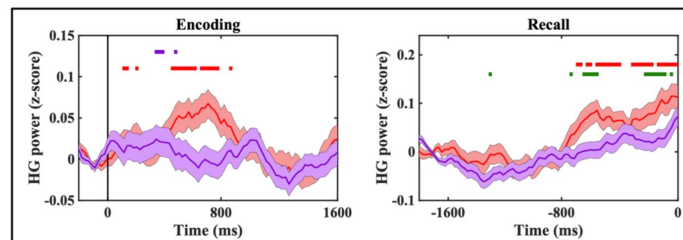
iEEG evoked response for AI (red) and mPFC (cyan) in the four experiments.

Green horizontal lines denote time periods where high-gamma power between the AI and mPFC were significantly different from each other. Red and cyan horizontal lines denote increase of high-gamma power compared to the resting baseline in the AI and mPFC respectively.

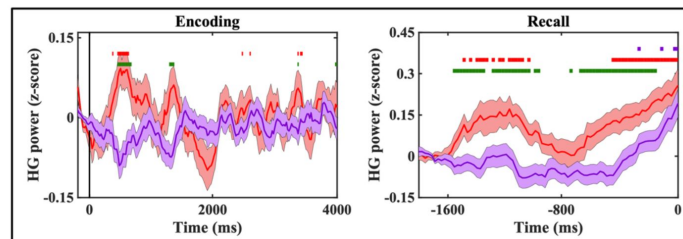
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

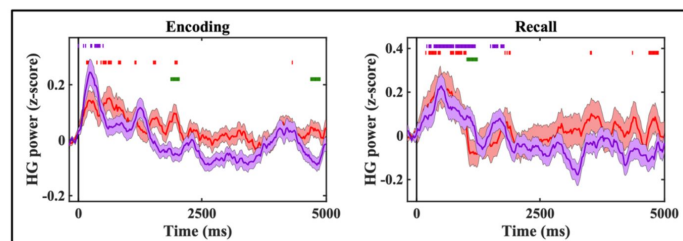
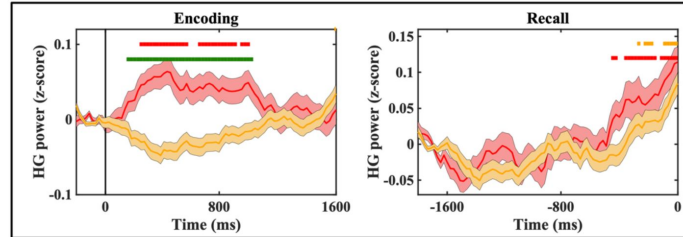


Figure S2.

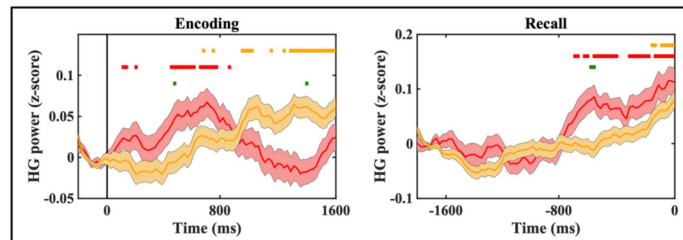
iEEG evoked response for AI (red) and dPPC (purple) in the four experiments.

Green horizontal lines denote time periods where high-gamma power between the AI and dPPC were significantly different from each other. Red and purple horizontal lines denote increase of high-gamma power compared to the resting baseline in the AI and dPPC respectively.

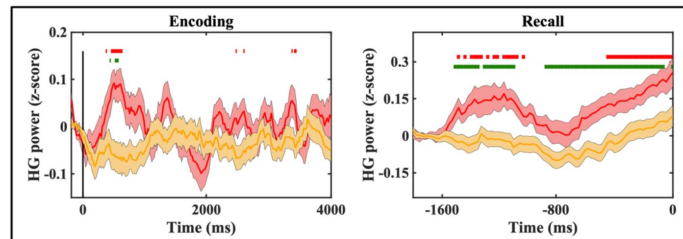
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

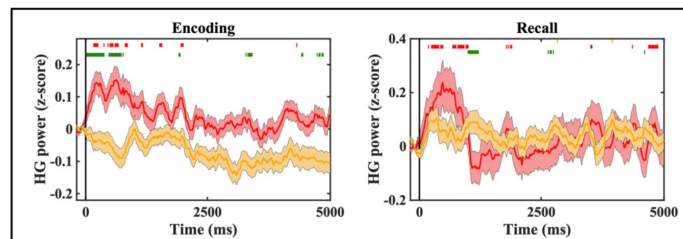
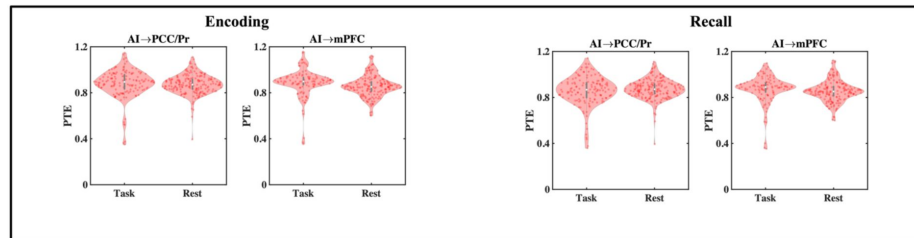


Figure S3.

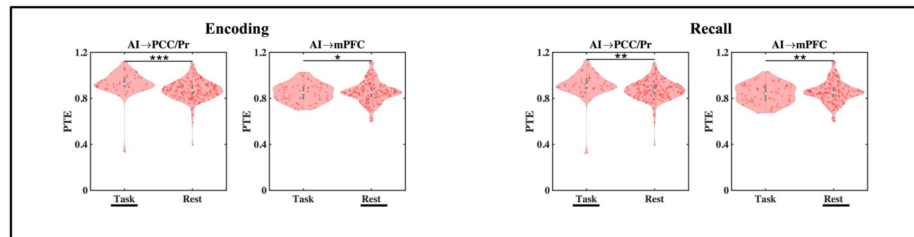
iEEG evoked response for AI (red) and MFG (orange) in the four experiments.

Green horizontal lines denote time periods where high-gamma power between the AI and MFG were significantly different from each other. Red and orange horizontal lines denote increase of high-gamma power compared to the resting baseline in the AI and MFG respectively.

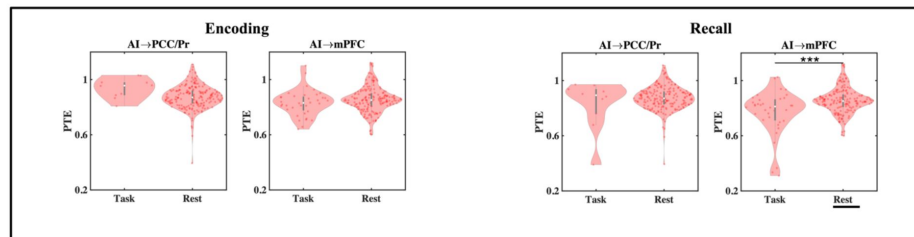
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

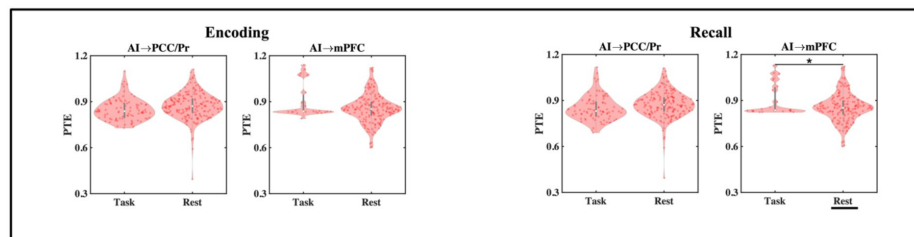
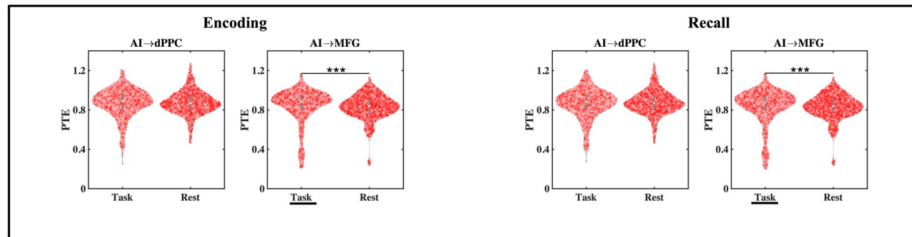


Figure S4.

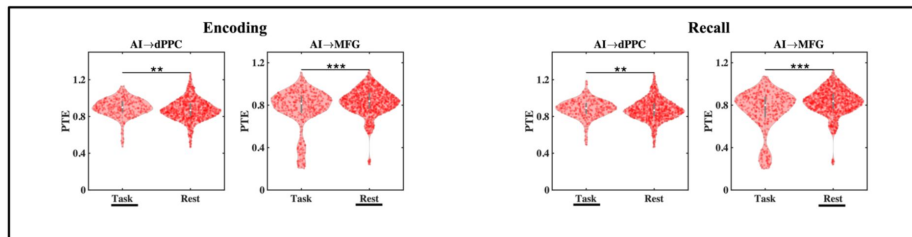
Differential directed information flow from the anterior insula to the DMN nodes during task versus resting-state, in broadband.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

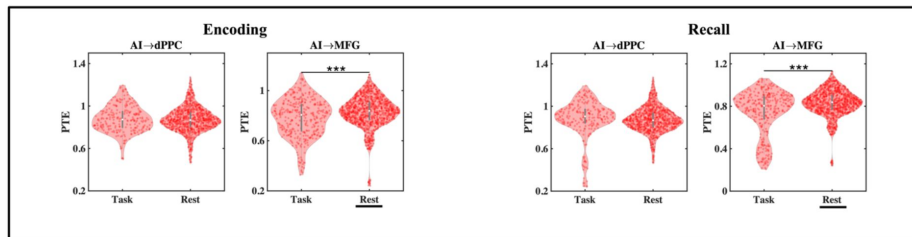
(a) Experiment 1 — VFR



(b) Experiment 2 — CATVFR



(c) Experiment 3 — PALVCR



(d) Experiment 4 — WMSM

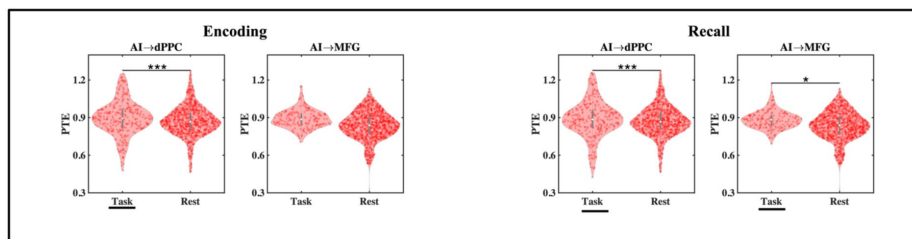


Figure S5.

Differential directed information flow from the anterior insula to the FPN nodes during task versus resting-state, in broadband.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Supplementary Results

Causal information flow from the AI to the DMN during encoding and recall in the CATVFR task

Encoding

Directed information flow from the AI to the PCC/precuneus was higher than the reverse, during memory encoding ($F(1, 84) = 36.18, p < 0.001$, Cohen's $d = 1.32$) (Figure 4b [↗](#)).

Recall

Directed information flow from the AI to the PCC/precuneus ($F(1, 83) = 29.54, p < 0.001$, Cohen's $d = 1.19$) was higher, than the reverse, during memory recall (Figure 4b [↗](#)).

These results demonstrate that the AI has strong causal information flow to the PCC/precuneus node of the DMN during both the encoding and recall phases of the CATVFR episodic memory task.

Causal information flow from the AI to the DMN during encoding and recall in the PALVCR task

Encoding

Directed information flow from the AI to the PCC/precuneus ($F(1, 17) = 22.19, p < 0.001$, Cohen's $d = 2.28$) was higher, than the reverse (Figure 4c [↗](#)).

Recall

Directed information flow from the AI to the PCC/precuneus was higher, than the reverse ($F(1, 17) = 6.45, p < 0.05$, Cohen's $d = 1.23$) (Figure 4c [↗](#)).

These results demonstrate that the AI has stronger causal information flow to the PCC/precuneus node of the DMN during both the encoding and recall phases of the PALVCR episodic memory task.

Causal information flow from the AI to the DMN during encoding and recall in the WMSM task

Encoding

Directed information flow from the AI to the PCC/precuneus ($F(1, 176) = 51.14, p < 0.001$, Cohen's $d = 1.08$) and mPFC ($F(1, 41) = 44.53, p < 0.001$, Cohen's $d = 2.08$) were higher, than the reverse (Figure 4d [↗](#)).

Recall

Directed information flow from the AI to the PCC/precuneus ($F(1, 177) = 36.86, p < 0.001$, Cohen's $d = 0.91$) and the mPFC ($F(1, 41) = 39.62, p < 0.001$, Cohen's $d = 1.96$) were also higher, than the reverse (Figure 4d [↗](#)).

These results demonstrate that the AI has stronger causal information flow to the PCC/precuneus and mPFC nodes of the DMN during both the encoding and recall phases of the WMSM task.

Causal information flow from AI to FPN nodes in the CATVFR task

We next examined directed information flow between the AI and FPN nodes during the categorized verbal free recall task.

Encoding

Directed information flow from the AI to the dPPC was higher, than the reverse ($F(1, 639) = 27.16$, $p < 0.001$, Cohen's $d = 0.41$) (Figure 5b [↗](#)).

Recall

Directed information flow from the AI to the dPPC was higher, than the reverse ($F(1, 639) = 20.48$, $p < 0.001$, Cohen's $d = 0.36$) (Figure 5b [↗](#)).

These results demonstrate that the AI has stronger causal information flow to the dPPC node of the FPN during both the encoding and recall phases of the CATVFR episodic memory task.

Causal information flow from AI to FPN nodes in the PALVCR task

We next examined directed information flow between the AI and FPN nodes during the paired associates learning verbal cued recall task.

Encoding

Directed information flow from the AI to the dPPC ($F(1, 476) = 38.25$, $p < 0.001$, Cohen's $d = 0.57$) was higher, than the reverse (Figure 5c [↗](#)).

Recall

Directed information flow from the AI to the dPPC ($F(1, 475) = 60.09$, $p < 0.001$, Cohen's $d = 0.71$) and MFG ($F(1, 709) = 9.90$, $p < 0.01$, Cohen's $d = 0.24$) were higher, than the reverse (Figure 5c [↗](#)).

These results demonstrate that the AI has stronger causal information flow to the dPPC node of the FPN during encoding and both dPPC and MFG nodes of the FPN during the recall phase of the PALVCR episodic memory task.

Causal information flow from the AI to FPN nodes in the WMSM task

Encoding

Directed information flow from the AI to the MFG ($F(1, 343) = 74.38$, $p < 0.001$, Cohen's $d = 0.93$) was higher, than the reverse (Figure 5d [↗](#)).

Recall

Directed information flow from the AI to the MFG ($F(1, 344) = 102.18$, $p < 0.001$, Cohen's $d = 1.09$) was higher, than the reverse (Figure 5d [↗](#)).

These results demonstrate that the AI has stronger causal information flow to the MFG node of the FPN during both the encoding and recall phases of the WMSM task.

Causal outflow hub during encoding and recall in the CATVFR task

Encoding

Net causal outflow from the AI is positive and higher than both PCC/precuneus ($F(1, 2023) = 59.97$, $p < 0.001$, Cohen's $d = 0.34$) and mPFC ($F(1, 2676) = 23.16$, $p < 0.001$, Cohen's $d = 0.19$) during encoding (**Figure 6b** [↗](#)).

We also found that the net causal outflow from the AI is higher than the MFG during encoding ($F(1, 3974) = 11.61$, $p < 0.001$, Cohen's $d = 0.11$) (**Figure 6b** [↗](#)). However, the net causal outflow from the AI was lower than the dPPC during encoding ($F(1, 3535) = 6.04$, $p < 0.05$, Cohen's $d = 0.08$) (**Figure 6b** [↗](#)).

Recall

Net causal outflow from the AI is positive and higher than both PCC/precuneus ($F(1, 1827) = 33.55$, $p < 0.001$, Cohen's $d = 0.27$) and mPFC ($F(1, 2656) = 29.81$, $p < 0.001$, Cohen's $d = 0.21$) during the recall phase of the categorized verbal free recall task (**Figure 6b** [↗](#)).

We also found that the net causal outflow from the AI is higher than the MFG during recall ($F(1, 3827) = 6.87$, $p < 0.01$, Cohen's $d = 0.08$) (**Figure 6b** [↗](#)).

Causal outflow hub during encoding and recall in the PALVCR task

Encoding

We found similar results for the paired associates learning verbal cued recall task where, net causal outflow from the AI is positive and higher than both PCC/precuneus ($F(1, 736) = 9.84$, $p < 0.01$, Cohen's $d = 0.23$) and mPFC ($F(1, 1079) = 21.93$, $p < 0.001$, Cohen's $d = 0.29$) during memory encoding (**Figure 6c** [↗](#)).

We also found that the net causal outflow from the AI is higher than the MFG during encoding ($F(1, 1779) = 14.45$, $p < 0.001$, Cohen's $d = 0.18$) (**Figure 6c** [↗](#)). However, the net causal outflow from the AI is lower than the dPPC during encoding ($F(1, 1261) = 8.72$, $p < 0.01$, Cohen's $d = 0.17$) (**Figure 6c** [↗](#)).

Recall

Net causal outflow from the AI is positive and higher than both PCC/precuneus ($F(1, 530) = 10.96$, $p < 0.001$, Cohen's $d = 0.29$) and mPFC ($F(1, 909) = 8.42$, $p < 0.01$, Cohen's $d = 0.19$) during memory recall (**Figure 6c** [↗](#)).

Net causal outflow from the AI is higher than both the dPPC ($F(1, 1041) = 31.15$, $p < 0.001$, Cohen's $d = 0.35$) and MFG ($F(1, 736) = 70.08$, $p < 0.001$, Cohen's $d = 0.62$) nodes of the FPN during the recall phase of the paired associates learning verbal cued recall task (**Figure 6c** [↗](#)).

Together, these results demonstrate that the AI is a causal outflow hub in its interactions with the PCC/precuneus and mPFC nodes of the DMN and the MFG node of the FPN, during both verbal memory encoding and recall.

Causal outflow hub during encoding and recall in the WMSM task

We next repeated our causal hub analysis during the encoding and recall phases of the water maze spatial memory task.

Encoding

We found that net causal outflow from the AI is positive and higher than both the PCC/precuneus ($F(1, 1669) = 168.5$, $p < 0.001$, Cohen's $d = 0.64$) and mPFC ($F(1, 1278) = 9.91$, $p < 0.01$, Cohen's $d = 0.18$) nodes of the DMN during encoding (**Figure 6d** [↗](#)).

We also found that net causal outflow from the AI is higher than both the dPPC ($F(1, 2501) = 7.10$, $p < 0.01$, Cohen's $d = 0.11$) and MFG ($F(1, 1977) = 73.49$, $p < 0.001$, Cohen's $d = 0.39$) nodes of the FPN during encoding (**Figure 6d** [↗](#)).

Recall

Net causal outflow from the AI is positive and higher than both the PCC/precuneus ($F(1, 1672) = 166.95$, $p < 0.001$, Cohen's $d = 0.63$) and mPFC ($F(1, 1270) = 12.75$, $p < 0.001$, Cohen's $d = 0.20$) nodes of the DMN during recall (**Figure 6d** [↗](#)).

Net causal outflow from the AI is also higher than the MFG ($F(1, 1985) = 90.81$, $p < 0.001$, Cohen's $d = 0.43$) node of the FPN during recall (**Figure 6d** [↗](#)).

Together, these results demonstrate that the AI is a causal outflow hub in its interactions with the PCC/precuneus and mPFC nodes of the DMN and also the dPPC and MFG nodes of the FPN, during both spatial memory encoding and recall.

Supplementary Tables

Participant ID	Gender	Age
001	F	48
002	F	49
003	F	39
006	F	20
010	F	30
014	F	47
015	F	54
018	M	47
019	F	34
020	F	48
021	M	38
022	M	24
023	M	32
024	F	36
025	F	19
026	F	24
027	M	48
028	F	27
029	F	33
030	M	23
032	F	19
033	F	31
034	F	29
035	F	45
036	M	49
039	F	28
041	M	34
042	F	28
044	M	58
045	M	51
049	F	52
050	M	20
051	F	24
052	F	19
053	F	39
054	M	23
056	M	34
057	M	53

Table S1.

Participant demographic information (total 177 participants).

059	F	44
060	F	36
062	F	23
063	M	23
064	M	56
065	F	34
066	M	39
067	F	45
068	F	39
069	M	26
070	F	40
074	M	24
075	M	50
076	M	29
077	F	47
078	F	22
080	F	43
081	F	33
082	M	39
084	M	25
087	M	51
089	M	36
090	F	52
091	M	28
092	M	44
093	M	24
094	M	47
095	F	35
097	M	34
098	F	38
100	F	43
101	F	26
102	M	34
105	M	25
106	M	26
107	M	25
108	F	23
109	F	43
111	M	20
114	F	31
115	M	47
118	M	33
119	F	26

Table S1. (continued)

120	F	33
121	M	34
123	F	29
124	F	40
125	F	44
127	F	40
128	M	26
129	F	34
130	M	57
131	M	24
134	M	64
135	M	47
136	F	16
137	F	21
138	M	41
141	F	44
142	F	43
144	M	53
147	M	47
148	F	59
149	F	28
150	F	49
151	M	36
153	M	38
155	M	37
156	M	27
157	M	22
158	F	45
159	F	42
161	F	53
162	F	30
163	M	45
164	M	37
166	M	38
167	M	33
168	M	24
171	M	36
172	F	22
173	F	18
174	M	29
175	M	34
176	F	41
177	F	23

Table S1. (continued)

Table S1. (continued)

178	M	40
180	F	21
181	M	22
184	M	42
186	M	27
187	F	51
189	M	22
190	F	57
193	M	37
195	M	44
196	M	18
200	M	25
202	F	29
203	F	36
204	F	25
207	F	39
212	M	46
221	M	57
222	F	20
223	F	42
227	M	32
228	F	58
230	F	56
232	M	27
234	M	25
236	F	51
238	M	27
239	M	27
240	F	37
245	M	30
247	F	61
251	M	31
260	F	57
263	M	30
264	F	52
268	F	32
271	M	37
274	F	44
275	M	41
276	M	28
279	F	57
283	F	29
284	F	32
286	F	57
292	F	39
297	M	24
298	F	24
299	M	43
302	M	48
303	F	62
304	F	33
310	M	20
312	M	21

Network pair	Number of electrode pairs (n)	Number of participants	Participant IDs (Gender/Age)
AI-PCC/Pr	142	18	030 (M/23), 049 (F/52), 054 (M/23), 057 (M/53), 062 (F/23), 114 (F/31), 115 (M/47), 134 (M/64), 153 (M/38), 158 (F/45), 168 (M/24), 193 (M/37), 196 (M/18), 204 (F/25), 236 (F/51), 240 (F/37), 286 (F/57), 299 (M/43)
AI-mPFC	112	20	026 (F/24), 027 (M/48), 049 (F/52), 057 (M/53), 062 (F/23), 114 (F/31), 115 (M/47), 123 (F/29), 153 (M/38), 163 (M/45), 168 (M/24), 189 (M/22), 193 (M/37), 196 (M/18), 204 (F/25), 223 (F/42), 228 (F/58), 247 (F/61), 274 (F/44), 299 (M/43)
AI-dPPC	586	28	030 (M/23), 032 (F/19), 033 (F/31), 049 (F/52), 054 (M/23), 057 (M/53), 062 (F/23), 065 (F/34), 080 (F/43), 114 (F/31), 115 (M/47), 128 (M/26), 134 (M/64), 153 (M/38), 158 (F/45), 163 (M/45), 168 (M/24), 173 (F/18), 189 (M/22), 193 (M/37), 196 (M/18), 204 (F/25), 232 (M/27), 236 (F/51), 240 (F/37), 247 (F/61), 286 (F/57), 299 (M/43)
AI-MFG	642	36	026 (F/24), 030 (M/23), 032 (F/19), 033 (F/31), 049 (F/52), 054 (M/23), 057 (M/53), 062 (F/23), 063 (M/23), 065 (F/34), 114 (F/31), 115 (M/47), 153 (M/38), 158 (F/45), 163 (M/45), 166 (M/38), 168 (M/24), 178 (M/40), 189 (M/22), 193 (M/37), 196 (M/18), 204 (F/25), 207 (F/39), 223 (F/42), 228 (F/58), 230 (F/56), 232 (M/27), 240 (F/37), 247 (F/61), 264 (F/52), 274 (F/44), 283 (F/29), 286 (F/57), 298 (F/24), 299 (M/43), 310 (M/20)

Table S2a.

Number of electrode pairs used in phase transfer entropy (PTE) analysis in the verbal free recall task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

Network pair	Number of electrode pairs (n)	Number of participants	Participant IDs (Gender/Age)
AI-PCC/Pr	46	7	114 (F/31), 141 (F/44), 158 (F/45), 204 (F/25), 240 (F/37), 245 (M/30), 286 (F/57)
AI-mPFC	64	12	026 (F/24), 114 (F/31), 141 (F/44), 163 (M/45), 189 (M/22), 204 (F/25), 228 (F/58), 245 (M/30), 247 (F/61), 271 (M/37), 274 (F/44), 303 (F/62)
AI-dPPC	327	14	028 (F/27), 032 (F/19), 065 (F/34), 114 (F/31), 141 (F/44), 158 (F/45), 163 (M/45), 189 (M/22), 204 (F/25), 240 (F/37), 245 (M/30), 247 (F/61), 271 (M/37), 286 (F/57)
AI-MFG	462	22	026 (F/24), 032 (F/19), 065 (F/34), 114 (F/31), 141 (F/44), 158 (F/45), 163 (M/45), 178 (M/40), 189 (M/22), 204 (F/25), 207 (F/39), 228 (F/58), 230 (F/56), 240 (F/37), 245 (M/30), 247 (F/61), 264 (F/52), 271 (M/37), 274 (F/44), 286 (F/57), 303 (F/62), 310 (M/20)

Table S2b.

Number of electrode pairs used in phase transfer entropy (PTE) analysis in the categorized verbal free recall task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

Table S2c.

Number of electrode pairs used in phase transfer entropy (PTE) analysis in the paired associates learning verbal cued recall task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

Network pair	Number of electrode pairs (n)	Number of participants	Participant IDs (Gender/Age)
AI-PCC/Pr	10	2	141 (F/44), 196 (M/18)
AI-mPFC	36	5	141 (F/44), 196 (M/18), 223 (F/42), 228 (F/58), 303 (F/62)
AI-dPPC	242	9	028 (F/27), 065 (F/34), 090 (F/52), 091 (M/28), 141 (F/44), 196 (M/18), 232 (M/27), 238 (M/27), 312 (M/21)
AI-MFG	362	14	065 (F/34), 090 (F/52), 091 (M/28), 141 (F/44), 196 (M/18), 207 (F/39), 223 (F/42), 228 (F/58), 230 (F/56), 232 (M/27), 238 (M/27), 283 (F/29), 303 (F/62), 312 (M/21)

Table S2d.

Number of electrode pairs used in phase transfer entropy (PTE) analysis in the water maze spatial memory task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

Network pair	Number of electrode pairs (n)	Number of participants	Participant IDs (Gender/Age)
AI-PCC/Pr	91	6	030 (M/23), 049 (F/52), 054 (M/23), 062 (F/23), 114 (F/31), 124 (F/40)
AI-mPFC	23	5	026 (F/24), 049 (F/52), 052 (F/19), 062 (F/23), 114 (F/31)
AI-dPPC	302	10	030 (M/23), 032 (F/19), 033 (F/31), 049 (F/52), 052 (F/19), 054 (M/23), 062 (F/23), 065 (F/34), 114 (F/31), 124 (F/40)
AI-MFG	177	10	026 (F/24), 030 (M/23), 032 (F/19), 033 (F/31), 049 (F/52), 052 (F/19), 054 (M/23), 062 (F/23), 065 (F/34), 114 (F/31)

Brain regions	Number of electrodes (n)	Number of participants	Participant IDs (Gender/Age)
AI	148	44	026 (F/24), 027 (M/48), 030 (M/23), 032 (F/19), 033 (F/31), 049 (F/52), 054 (M/23), 057 (M/53), 062 (F/23), 063 (M/23), 065 (F/34), 080 (F/43), 114 (F/31), 115 (M/47), 123 (F/29), 128 (M/26), 134 (M/64), 150 (F/49), 153 (M/38), 158 (F/45), 163 (M/45), 166 (M/38), 168 (M/24), 173 (F/18), 178 (M/40), 189 (M/22), 193 (M/37), 196 (M/18), 204 (F/25), 207 (F/39), 223 (F/42), 228 (F/58), 230 (F/56), 232 (M/27), 236 (F/51), 240 (F/37),

Table S3a.

Number of electrodes in each node used in high-gamma power analysis in the verbal free recall task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

			247 (F/61), 264 (F/52), 274 (F/44), 283 (F/29), 286 (F/57), 298 (F/24), 299 (M/43), 310 (M/20)
PCC/Pr	143	47	006 (F/20), 010 (F/30), 015 (F/54), 018 (M/47), 023 (M/32), 030 (M/23), 034 (F/29), 039 (F/28), 044 (M/58), 049 (F/52), 051 (F/24), 054 (M/23), 057 (M/53), 062 (F/23), 070 (F/40), 074 (M/24), 076 (M/29), 077 (F/47), 081 (F/33), 084 (M/25), 094 (M/47), 101 (F/26), 105 (M/25), 106 (M/26), 114 (F/31), 115 (M/47), 134 (M/64), 135 (M/47), 138 (M/41), 153 (M/38), 155 (M/37), 158 (F/45), 162 (F/30), 168 (M/24), 175 (M/34), 186 (M/27), 193 (M/37), 196 (M/18), 203 (F/36), 204 (F/25), 236 (F/51), 240 (F/37), 268 (F/32), 275 (M/41), 286 (F/57), 297 (M/24), 299 (M/43)
mPFC	312	55	018 (M/47), 022 (M/24), 026 (F/24), 027 (M/48), 034 (F/29), 036 (M/49), 039 (F/28), 049 (F/52), 051 (F/24), 053 (F/39), 056 (M/34), 057 (M/53), 059 (F/44), 060 (F/36), 062 (F/23), 070 (F/40), 074 (M/24), 075 (M/50), 077 (F/47), 081 (F/33), 084 (M/25), 098 (F/38), 106 (M/26), 114 (F/31), 115 (M/47), 121 (M/34), 123 (F/29), 129 (F/34), 130 (M/57), 131 (M/24), 142 (F/43), 151 (M/36), 153 (M/38), 155 (M/37), 156 (M/27), 163 (M/45), 167 (M/33), 168 (M/24), 175 (M/34), 187 (F/51), 189 (M/22), 193 (M/37), 196 (M/18), 200 (M/25), 202 (F/29), 203 (F/36), 204 (F/25), 222 (F/20), 223 (F/42), 228 (F/58), 247 (F/61), 274 (F/44), 275 (M/41), 299 (M/43), 304 (F/33)
dPPC	537	89	001 (F/48), 003 (F/39), 006 (F/20), 010 (F/30), 015 (F/54), 018 (M/47), 020 (F/48), 023 (M/32), 030 (M/23), 032 (F/19), 033 (F/31), 035 (F/45), 036 (M/49), 039 (F/28), 042 (F/28), 044 (M/58), 049 (F/52), 050 (M/20), 053 (F/39), 054 (M/23), 056 (M/34), 057 (M/53), 059 (F/44), 062 (F/23),

Table S3a. (continued)

Table S3a. (continued)

			065 (F/34), 066 (M/39), 067 (F/45), 068 (F/39), 069 (M/26), 070 (F/40), 074 (M/24), 075 (M/50), 077 (F/47), 080 (F/43), 084 (M/25), 089 (M/36), 094 (M/47), 101 (F/26), 102 (M/34), 105 (M/25), 106 (M/26), 111 (M/20), 114 (F/31), 115 (M/47), 120 (F/33), 121 (M/34), 125 (F/44), 128 (M/26), 130 (M/57), 134 (M/64), 135 (M/47), 138 (M/41), 147 (M/47), 151 (M/36), 153 (M/38), 156 (M/27), 158 (F/45), 161 (F/53), 162 (F/30), 163 (M/45), 164 (M/37), 168 (M/24), 171 (M/36), 173 (F/18), 174 (M/29), 175 (M/34), 176 (F/41), 177 (F/23), 184 (M/42), 186 (M/27), 189 (M/22), 193 (M/37), 195 (M/44), 196 (M/18), 203 (F/36), 204 (F/25), 232 (M/27), 234 (M/25), 236 (F/51), 240 (F/37), 247 (F/61), 251 (M/31), 260 (F/57), 268 (F/32), 275 (M/41), 286 (F/57), 292 (F/39), 297 (M/24), 299 (M/43)
MFG	538	97	002 (F/49), 003 (F/39), 006 (F/20), 015 (F/54), 020 (F/48), 022 (M/24), 023 (M/32), 026 (F/24), 030 (M/23), 032 (F/19), 033 (F/31), 034 (F/29), 036 (M/49), 039 (F/28), 042 (F/28), 045 (M/51), 049 (F/52), 051 (F/24), 053 (F/39), 054 (M/23), 056 (M/34), 057 (M/53), 059 (F/44), 060 (F/36), 062 (F/23), 063 (M/23), 065 (F/34), 066 (M/39), 067 (F/45), 069 (M/26), 070 (F/40), 074 (M/24), 075 (M/50), 076 (M/29), 077 (F/47), 081 (F/33), 084 (M/25), 089 (M/36), 098 (F/38), 102 (M/34), 105 (M/25), 106 (M/26), 114 (F/31), 115 (M/47), 121 (M/34), 127 (F/40), 129 (F/34), 130 (M/57), 131 (M/24), 135 (M/47), 136 (F/16), 137 (F/21), 142 (F/43), 147 (M/47), 148 (F/59), 149 (F/28), 151 (M/36), 153 (M/38), 155 (M/37), 156 (M/27), 158 (F/45), 159 (F/42), 162 (F/30), 163 (M/45), 164 (M/37), 166 (M/38), 168 (M/24), 172 (F/22), 175 (M/34), 177 (F/23), 178 (M/40), 186 (M/27),
			189 (M/22), 193 (M/37), 195 (M/44), 196 (M/18), 200 (M/25), 203 (F/36), 204 (F/25), 207 (F/39), 222 (F/20), 223 (F/42), 228 (F/58), 230 (F/56), 232 (M/27), 240 (F/37), 247 (F/61), 260 (F/57), 264 (F/52), 274 (F/44), 275 (M/41), 283 (F/29), 286 (F/57), 298 (F/24), 299 (M/43), 304 (F/33), 310 (M/20)

Brain regions	Number of electrodes (n)	Number of participants	Participant IDs (Gender/Age)
AI	107	25	026 (F/24), 028 (F/27), 032 (F/19), 065 (F/34), 114 (F/31), 141 (F/44), 158 (F/45), 163 (M/45), 178 (M/40), 189 (M/22), 204 (F/25), 207 (F/39), 228 (F/58), 230 (F/56), 236 (F/51), 239 (M/27), 240 (F/37), 245 (M/30), 247 (F/61), 264 (F/52), 271 (M/37), 274 (F/44), 286 (F/57), 303 (F/62), 310 (M/20)
PCC/Pr	74	21	015 (F/54), 039 (F/28), 041 (M/34), 044 (M/58), 074 (M/24), 094 (M/47), 105 (M/25), 106 (M/26), 114 (F/31), 135 (M/47), 141 (F/44), 157 (M/22), 158 (F/45), 186 (M/27), 204 (F/25), 227 (M/32), 236 (F/51), 240 (F/37), 245 (M/30), 275 (M/41), 286 (F/57)
mPFC	116	33	026 (F/24), 029 (F/33), 036 (M/49), 039 (F/28), 041 (M/34), 056 (M/34), 060 (F/36), 074 (M/24), 075 (M/50), 106 (M/26), 107 (M/25), 114 (F/31), 119 (F/26), 130 (M/57), 131 (M/24), 141 (F/44), 163 (M/45), 167 (M/33), 180 (F/21), 181 (M/22), 187 (F/51), 189 (M/22), 202 (F/29), 204 (F/25), 212 (M/46), 222 (F/20), 228 (F/58), 245 (M/30), 247 (F/61), 271 (M/37), 274 (F/44), 275 (M/41), 303 (F/62)

Table S3b.

Number of electrodes in each node used in high-gamma power analysis in the categorized verbal free recall task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

dPPC	357	57	015 (F/54), 028 (F/27), 032 (F/19), 035 (F/45), 036 (M/49), 039 (F/28), 042 (F/28), 044 (M/58), 050 (M/20), 056 (M/34), 065 (F/34), 066 (M/39), 067 (F/45), 069 (M/26), 074 (M/24), 075 (M/50), 089 (M/36), 092 (M/44), 094 (M/47), 102 (M/34), 105 (M/25), 106 (M/26), 108 (F/23), 111 (M/20), 114 (F/31), 119 (F/26), 130 (M/57), 135 (M/47), 141 (F/44), 144 (M/53), 147 (M/47), 157 (M/22), 158 (F/45), 163 (M/45), 171 (M/36), 174 (M/29), 176 (F/41), 181 (M/22), 184 (M/42), 186 (M/27), 189 (M/22), 190 (F/57), 204 (F/25), 212 (M/46), 221 (M/57), 227 (M/32), 236 (F/51), 240 (F/37), 245 (M/30), 247 (F/61), 251 (M/31), 260 (F/57), 271 (M/37), 275 (M/41), 279 (F/57), 286 (F/57), 302 (M/48)
MFG	375	58	015 (F/54), 021 (M/38), 026 (F/24), 029 (F/33), 032 (F/19), 036 (M/49), 039 (F/28), 041 (M/34), 042 (F/28), 045 (M/51), 056 (M/34), 060 (F/36), 065 (F/34), 066 (M/39), 067 (F/45), 069 (M/26), 074 (M/24), 075 (M/50), 089 (M/36), 092 (M/44), 093 (M/24), 102 (M/34), 105 (M/25), 106 (M/26), 107 (M/25), 108 (F/23), 114 (F/31), 119 (F/26), 130 (M/57), 131 (M/24), 135 (M/47), 141 (F/44), 147 (M/47), 157 (M/22), 158 (F/45), 163 (M/45), 178 (M/40), 181 (M/22), 186 (M/27), 189 (M/22), 204 (F/25), 207 (F/39), 212 (M/46), 221 (M/57), 222 (F/20), 228 (F/58), 230 (F/56), 240 (F/37), 245 (M/30), 247 (F/61), 260 (F/57), 264 (F/52), 271 (M/37), 274 (F/44), 275 (M/41), 286 (F/57), 303 (F/62), 310 (M/20)

Table S3b. (continued)

Brain regions	Number of electrodes (n)	Number of participants	Participant IDs (Gender/Age)
AI	84	15	028 (F/27), 065 (F/34), 090 (F/52), 091 (M/28), 141 (F/44), 196 (M/18), 207 (F/39), 223 (F/42), 228 (F/58), 230 (F/56), 232 (M/27), 238 (M/27), 283 (F/29), 303 (F/62), 312 (M/21)
PCC/Pr	28	10	023 (M/32), 074 (M/24), 078 (F/2), 106 (M/26), 141 (F/44), 162 (F/30), 175 (M/34), 196 (M/18), 284 (F/32), 297 (M/24)
mPFC	78	20	036 (M/49), 056 (M/34), 060 (F/36), 074 (M/24), 082 (M/39), 097 (M/34), 106 (M/26), 121 (M/34), 130 (M/57), 131 (M/24), 141 (F/44), 142 (F/43), 175 (M/34), 196 (M/18), 202 (F/29), 212 (M/46), 223 (F/42), 228 (F/58), 263 (M/30), 303 (F/62)
dPPC	192	39	001 (F/48), 003 (F/39), 023 (M/32), 028 (F/27), 035 (F/45), 036 (M/49), 042 (F/28), 050 (M/20), 056 (M/34), 065 (F/34), 066 (M/39), 069 (M/26), 074 (M/24), 078 (F/22), 082 (M/39), 087 (M/51), 089 (M/36), 090 (F/52), 091 (M/28), 095 (F/35), 097 (M/34), 102 (M/34), 106 (M/26), 109 (F/43), 111 (M/20), 118 (M/33), 121 (M/34), 130 (M/57), 141 (F/44), 162 (F/30), 175 (M/34), 196 (M/18), 212 (M/46), 232 (M/27), 238 (M/27), 276 (M/28), 284 (F/32), 297 (M/24), 312 (M/21)
MFG	204	44	002 (F/49), 003 (F/39), 023 (M/32), 036 (M/49), 042 (F/28), 056 (M/34), 060 (F/36), 065 (F/34), 066 (M/39), 069 (M/26), 074 (M/24), 078 (F/22), 082 (M/39), 089 (M/36), 090 (F/52), 091 (M/28), 095 (F/35), 097 (M/34), 100 (F/43), 102 (M/34), 106 (M/26), 118 (M/33), 121 (M/34), 130 (M/57), 131 (M/24), 136 (F/16), 141 (F/44), 142 (F/43), 149 (F/28), 162 (F/30), 175 (M/34), 196 (M/18), 207 (F/39), 212 (M/46), 223 (F/42), 228 (F/58), 230 (F/56), 232 (M/27), 238 (M/27), 263 (M/30), 276 (M/28), 283 (F/29), 303 (F/62), 312 (M/21)

Table S3c.

Number of electrodes in each node used in high-gamma power analysis in the paired associates learning verbal cued recall task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

Table S3d.

Number of electrodes in each node used in power spectral density (PSD) analysis in the water maze spatial memory task.

AI: anterior insula, PCC: posterior cingulate cortex, Pr: precuneus, mPFC: medial prefrontal cortex, dPPC: dorsal posterior parietal cortex, MFG: middle frontal gyrus.

Brain regions	Number of electrodes (n)	Number of participants	Participant IDs (Gender/Age)
AI	59	11	026 (F/24), 030 (M/23), 032 (F/19), 033 (F/31), 049 (F/52), 052 (F/19), 054 (M/23), 062 (F/23), 065 (F/34), 114 (F/31), 124 (F/40)
PCC/Pr	89	21	006 (F/20), 010 (F/30), 015 (F/54), 018 (M/47), 023 (M/32), 024 (F/36), 030 (M/23), 034 (F/29), 041 (M/34), 044 (M/58), 049 (F/52), 051 (F/24), 054 (M/23), 062 (F/23), 064 (M/56), 074 (M/24), 077 (F/47), 101 (F/26), 106 (M/26), 114 (F/31), 124 (F/40)
mPFC	77	17	014 (F/47), 018 (M/47), 025 (F/19), 026 (F/24), 034 (F/29), 041 (M/34), 049 (F/52), 051 (F/24), 052 (F/19), 056 (M/34), 060 (F/36), 062 (F/23), 074 (M/24), 075 (M/50), 077 (F/47), 106 (M/26), 114 (F/31)
dPPC	226	36	001 (F/48), 006 (F/20), 010 (F/30), 014 (F/47), 015 (F/54), 018 (M/47), 019 (F/34), 023 (M/32), 024 (F/36), 025 (F/19), 030 (M/23), 032 (F/19), 033 (F/31), 042 (F/28), 044 (M/58), 049 (F/52), 050 (M/20), 052 (F/19), 054 (M/23), 056 (M/34), 062 (F/23), 064 (M/56), 065 (F/34), 066 (M/39), 067 (F/45), 068 (F/39), 069 (M/26), 074 (M/24), 075 (M/50), 077 (F/47), 089 (M/36), 101 (F/26), 106 (M/26), 114 (F/31), 124 (F/40), 177 (F/23)
MFG	147	33	006 (F/20), 014 (F/47), 015 (F/54), 019 (F/34), 021 (M/38), 023 (M/32), 025 (F/19), 026 (F/24), 030 (M/23), 032 (F/19), 033 (F/31), 034 (F/29), 041 (M/34), 042 (F/28), 045 (M/51), 049 (F/52), 051 (F/24), 052 (F/19), 054 (M/23), 056 (M/34), 060 (F/36), 062 (F/23), 065 (F/34), 066 (M/39), 067 (F/45), 069 (M/26), 074 (M/24), 075 (M/50), 077 (F/47), 089 (M/36), 106 (M/26), 114 (F/31), 177 (F/23)

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Reviewer #1 (Public Review):

Summary:

Das and Menon describe an analysis of a large open-source iEEG dataset (UPENN-RAM). From encoding and recall phases of memory tasks, they analyzed power and phase-transfer entropy as a measure of directed information flow in regions across a hypothesized tripartite network system. The anterior insula (AI) was found to have heightened high gamma power

during encoding and retrieval, which corresponded to suppression of high gamma power in the posterior cingulate cortex (PCC) during encoding but not recall. In contrast, directed information flow from (but not to) AI to mPFC/PCC and dorsal posterior parietal/middle frontal cortex is high during both time periods when PTE is analyzed with broadband but not narrowband activity. They claim that these findings significantly advance an understanding of how network communication facilitates cognitive operations such as control over memory and that the AI of the salience network (SN) is responsible for governing the switch between the frontoparietal network (FPN) and default-mode network (DMN) when shifting between externally- and internally-driven processing.

I find this question interesting and important and agree with the authors that iEEG presents a unique opportunity to investigate the temporal dynamics within network nodes. However, I am not convinced that their claims are supported by the results currently presented. In particular, the fact that network-level communication is not modulated significantly compared to rest and does not relate to behavior suggests that PTE analyses may not be tapping into task-relevant communication. Moreover, dissociation of network effects - present during both encoding and recall - from local power suppression effects - present only during encoding - suggests that these sets of results may index separate and not unitary task processes.

Strengths:

- The authors present results from an impressively sized iEEG sample. For reader context, this type of invasive human data is difficult and time-consuming to collect and many similar studies in high-level journals include 5-20 participants, typically not all of whom have electrodes in all regions of interest. It is excellent that they have been able to leverage open-source data in this way.
- Preprocessing of iEEG data also seems sensible and appropriate based on field standards.
- The authors tackle the replication issues inherent in much of the literature by replicating findings across task contexts, demonstrating that the principles of network communication evidenced by their results generalize in multiple task memory contexts. Again, the number of iEEG patients who have multiple tasks' worth of data is impressive.

Weaknesses:

- The motivation for investigating the tripartite network during memory tasks is not currently well-elaborated. Though the authors mention, for example, that "the formation of episodic memories relies on the intricate interplay between large-scale brain networks (p. 4)", there are no citations provided for this statement, and the reader is unable to evaluate whether the nodes and networks evidenced to support these processes are the same as networks measured here.
- In addition, though the tripartite network has been proposed to support cognitive control processes, and the neural basis of cognitive control is the framed focus of this work, the authors do not demonstrate that they have measured cognitive control in addition to simple memory encoding and retrieval processes. Tasks that have investigated cognitive control over memory (such as those cited on p. 13 - Badre et al., 2005; Badre & Wagner, 2007; Wagner et al., 2001; Wagner et al., 2005) generally do not simply include encoding, delay, and recall (as the tasks used here), but tend to include some manipulation that requires participants to engage control processes over memory retrieval, such as task rules governing what choice should be made at recall (e.g., from Badre et al., 2005 Fig. 1: congruency of match, associative strength, number of choices, semantic similarity). Moreover, though there are task-responsive signatures in the nodes of the tripartite networks, concluding that cognitive control is present because cognitive control networks are active would be a reverse inference.

- It is currently unclear if the directed information flow from AI to DMN and FPN nodes truly arises from task-related processes such as cognitive control or if it is a function of static brain network characteristics constrained by anatomy (such as white matter connection patterns, etc.). This is a concern because the authors did not find that influences of AI on DMN or FPN are increased relative to a resting baseline (collected during the task) or that directed information flow differs in successful compared to unsuccessful retrieval. I doubt that this AI influence is 1) supporting a switch between the DMN and FPN via the SN or 2) relevant for behavior if it doesn't differ from baseline-active task or across accuracy conditions. An additional comparison that may help investigate whether this is reflective of static connectivity characteristics would be a baseline comparison during non-task rest or sleep periods.
- Related to the above concern, it is also questionable how directed information flow from AI facilitates switching between FPN and DMN during both encoding and recall if high gamma activity does not significantly differ in AI versus PCC or mPFC during recall as it does during encoding. It seems erroneous to conclude that the network-level communication is happening or happening with the same effect during both task time points when these effects are decoupled in such a way from the power findings.
- Missing information about the methods used for time-frequency conversion for power calculation and the power normalization/baseline-correction procedure bars a thorough evaluation of power calculation methods and results.

If revisions to the manuscript can address concerns about directed information flow possibly being due to anatomical constraints - such as by indicating that directed information flow is not present during non-task rest or sleep - this work may convey important information about the structure and order of communication between these networks during attention to tasks in general. However, the ability of the findings to address cognitive control-specific communication and the nature of neurophysiological mechanisms of this communication - as opposed to the temporal order and structure of recruited networks - may be limited.

Because phase-transfer entropy is presented as a "causal" analysis in this investigation (PTE), I also believe it is important to highlight for readers recent discussions surrounding the description of "causal mechanisms" in neuroscience (see "Confusion about causation" section from Ross and Bassett, 2024, *Nature Neuroscience*). A large proportion of neuroscientists (admittedly, myself included) use "causal" only to refer to a mechanism whose modulation or removal (with direct manipulation, such as by lesion or stimulation) is known to change or control a given outcome (such as a successful behavior). As Ross and Bassett highlight, it is debatable whether such mechanistic causality is captured by Granger "causality" (a.k.a. Granger prediction) or the parametric PTE, and the imprecise use of "causation" may be confusing. The authors could consider amending language regarding this analysis if they are concerned about bridging these definitions of causality across a wide audience.

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Reviewer #2 (Public Review):

In this study, the authors leverage a large public dataset of intracranial EEG (the University of Pennsylvania RAM repository) to examine electrophysiologic network dynamics involving the participation of salience, frontoparietal, and default mode networks in the completion of several episodic memory tasks. They do this through a focus on the anterior insula (AI; salience network), which they hypothesize may help switch engagement between the DMN and FPN in concert with task demands. By analyzing high-gamma spectral power and phase transfer entropy (PTE; a putative measure of information "flow"), they show that the AI

shows higher directed PTE towards nodes of both the DMN and FPN, during encoding and recall, across multiple tasks. They further demonstrate that high-gamma power in the PCC/precuneus is decreased relative to the AI during memory encoding. They interpret these results as evidence of "triple-network" control processes in memory tasks, governed by a key role of the AI.

I commend the authors on leveraging this large public dataset to help contextualize network models of brain function with electrophysiological mechanisms - a key problem in much of the fMRI literature. I also appreciate that the authors emphasized replicability across multiple memory tasks, in an effort to demonstrate conserved or fundamental mechanisms that support a diversity of cognitive processes. However, I believe that their strong claims regarding causal influences within circumscribed brain networks cannot be supported by the evidence as presented. In my efforts to clearly communicate these inadequacies, I will suggest several potential analyses for the authors to consider that might better link the data to their central hypotheses.

(1) As a general principle, the effects that the authors show - both in regards to their high-gamma power analysis and PTE analysis - do not offer sufficient specificity for a reader to understand whether these are general effects that may be repeated throughout the brain, or whether they reflect unique activity to the networks/regions that are laid out in the Introduction's hypothesis. This lack of specificity manifests in several ways, and is best communicated through examples of control analyses.

First, the PTE analysis is focused solely on the AI's interactions with nodes of the DMN and FPN; while it makes sense to focus on this putative "switch" region, the fact that the authors report significant PTE from the AI to nodes of both networks, in encoding and retrieval, across all tasks and (crucially) also at baseline, raises questions about the meaningfulness of this statistic. One way to address this concern would be to select a control region that would be expected to have little/no directed causal influence on these networks and repeat the analysis. Alternatively (or additionally), the authors could examine the time course of PTE as it evolves throughout an encoding/retrieval interval, and relate that to the timing of behavioral events or changes in high-gamma power. This would directly address an important idea raised in their own Discussion, "the AI is well-positioned to dynamically engage and disengage with other brain areas."

Second, the authors state that high-gamma suppression in the PCC/precuneus relative to the AI is an anatomically specific signature that is not present in the FPN. This claim does not seem to be supported by their own evidence as presented in the Supplemental Data (Figures S2 and S3), which to my eye show clear evidence of relative suppression in the MFG and dPPC (e.g. S2a and S3a, most notably) which are notated as "significant" with green bars. I appreciate that the magnitude of this effect may be greater in the PCC/precuneus, but if this is the claim it should be supported by appropriate statistics and interpretation.

(2) I commend the authors on emphasizing replicability, but I found their Bayes Factor (BF) analysis to be difficult to interpret and qualitatively inconsistent with the results that they show. For example, the authors state that BF analysis demonstrates "high replicability" of the gamma suppression effect in Figure 3a with that of 3c and 3d. While it does appear that significant effects exist across all three tasks, the temporal structure of high gamma signals appears markedly different between the two in ways that may be biologically meaningful. Moreover, it appears that the BF analysis did not support replicability between VFR and CATVFR, which is very surprising; these are essentially the same tasks (merely differing in the presence of word categories) and would be expected to have the highest degree of concordance, not the lowest. I would suggest the authors try to analytically or conceptually reconcile this surprising finding.

To aid in interpretability, it would be extremely helpful for the authors to assess across-task similarity in high-gamma power on a within-subject basis, which they are well-powered to do. For example, could they report the correlation coefficient between HGP timecourses in paired-associates versus free-recall tasks, to better establish whether these effects are consistent on a within-subject basis? This idea could similarly be extended to the PTE analysis. Across-subject correlations would also be a welcome analysis that may provide readers with better-contextualized effect sizes than the output of a Bayes Factor analysis.

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